Salience Theory and Risk Anomalies

Jiaqi Guo[†] & Youwei Li[‡]

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Keywords: Salience theory; choice under risk; risk-return trade-off; CAPM; retail investor trading

JEL classification: G11, G12, G14, G41

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Abstract

Investors' risk attitudes and demand for stocks is driven by their salient returns (Bordalo, Gennaioli, and Shleifer, 2012). For high-risk stocks prone to realizing such salient payoffs, we hypothesize opposing impacts on the risk-return relation contingent on salience of upsides versus downsides. Using various risk measures, we show that the risk-return relation is significantly positive for stocks with salient downsides but negative for stocks with salient upsides. We provide evidence that retail investors exhibit strong preference for stocks with salient payoffs, and demand more (less) high-risk stocks in the presence of salient upsides (downsides). Our findings hold after controlling for alternative explanations including lottery preference, arbitrage risk, reference-dependent preference, disagreement and coskewness risk. Our findings contribute to reconcile the conflicting empirical evidence on risk-return patterns.

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1. Introduction

Salient theory of Bordalo, Gennaioli and Shleifer (2012) (thereafter, BGS (2012)) extends the standard expected utility theory and the prospect theory of choice under risk. It assumes that investors' attention is drawn to returns which are most different or salient relative to the average. Intuitively, investors pay attention to and overweight both upside and downside of a risk choice. When a stock's upside return is salient, investors are attracted to it and exhibit risk-seeking behavior. The excess demand results in overvaluation and lower subsequent returns. Conversely, when the downside is salient, investors exhibit risk-averse behavior, resulting in undervaluation and higher subsequent returns.

In this paper, we explore how the salience theory of choice under risk can explain the puzzling empirical patterns in the risk-return relationship for stocks. While neoclassic theory predicts a positive relation between risk and expected return (e.g., CAPM of Sharpe (1964) and Lintner (1965); Merton (1987)), recent empirical studies show a striking weak or negative risk-return pattern. Starting with the early work by Black, Jensen and Scholes (1972), who document stock market beta is unrelated to expected return, a growing body of literature reports stock return actually declines with its beta (e.g., Frazzini and Pedersen, 2014; Schneider, Wagner and Zechner, 2020; and Novy-Marx and Velikov, 2022).

Our rationale is as follows. For any given stock, when its upside returns are salient and stand out, investors will overweight these salient upsides and exhibit risk-seeking behavior towards that stock. However, this risk-seeking effect is comparatively stronger for high-risk stocks versus low-risk stocks. This is because high-risk stocks, due to their higher volatility and wider return distributions, are more prone to realizing such extreme positive outlier returns that become salient upsides. Consequently, when high-risk stocks deliver salient upside gains, investors will find themselves exhibiting comparatively greater risk-seeking behavior towards these high-risk stocks versus lower-risk alternatives. This amplified risk-seeking behavior will drive greater overvaluation of the high-risk stocks, leading to a negative risk-return relationship among stocks with salient upsides.

On the other hand, for stocks with salient downsides, investors become comparatively more risk-averse for high-risk stocks that are likely to experience extreme negative outliers, thus undervaluing more these stocks versus low-risk stocks. This will result in a positive risk-return relationship among stocks with salient downsides Moreover, due to short-sale constraints, the negative relation for salient upside stocks is expected to be more pronounced than the positive relation for salient downside stocks.

We test our hypothesis by focusing on the empirical setting of the beta-return relation, with other risk-return settings as extensions. We start with this analysis for three main reasons. First, the CAPM provides an initial and fundamental framework to formalize the relationship between risk and expect return. Second, investors are most likely to use beta based on the CAPM to forecast future return. Graham and Harvey (2001) show that the CAPM is widely used by firms in the financial industry to measure cost of capital, while Berk and van Binsbergen (2016) and Barber, Huang, and Odean (2016) find that investors rely on the CAPM to form trading strategies. Third, the beta-return puzzle has been widely studied in the literature as least as early as Black, Jensen and Scholes (1972).[1] Using the salience measure in Cosemans

[1] Other studies include Tinic and West (1984), Fama and French, (1992), Cohen, Polk, and Vuolteenaho (2005), Baker, Bradley, and Wurgler (2011), Savor and Wilson (2014), Frazzini and Pedersen (2014), Jylha, (2018), Liu, Stambaugh, and Yuan (2018), Hendershott, Livdan, and Rösch (2020) and among others.

and Frehen (2021), we find strong evidence in support of the notion that the risk-return relation strictly depends on whether the stock payoff is salient upside or downside.

We start our empirical tests with a portfolio analysis. In each month, we double sort firms into quintile-by-quintile portfolios by beta and salience measure. The portfolios are held for one month. We benchmark the portfolio performance to the Fama and French's (2015) five-factor model. Consistent with many studies in the literature, there is no significant relation between beta and return. More importantly, consistent with our hypothesis, we find that the beta-return relation is strictly negative when the upside of stock return is salient but positive when the downside is salient. Specifically, among stocks with salient upsides, the alphas decrease monotonically from -0.269% within the low-beta group to -1.047% within the high-beta group, with the spread between high- and low-beta portfolios of -0.778% and *t*-statistic of -4.13. This suggests that stocks with salient upsides are all overvalued, but the overvaluation is much stronger for high-risk stocks compared to low-risk stocks.

In contrast, for stocks with salient downsides, all stocks earn positive alphas, and high beta stocks outperform low beta stocks by 0.465% with a *t*-statistic of 2.65. The results remain consistent in the regression analysis, even after controlling for various firm characteristics and survive a comprehensive list of robustness checks including alternative beta estimations, residual beta controlling for skewness, alternative salience measures, different stock exchanges, subperiods as well as exclusion of small and highly illiquid stocks.

Next, we examine the salience effect on the beta-return relation under different high and low market volatility states. We expect the salience effect to be stronger during high market volatility states than low volatility states because the stock's salient return tends to be more likely and stronger during volatile market states. Consistent with our conjecture, we find that the salience effect on the beta-return relation exists in both market states but tends to be much more pronounced in high market volatility states than low volatility states.

We then offer direct evidence of how salience-induced mispricing impacts the beta-return relationship from the perspective of retail investors, who are particularly susceptible to behavioral biases. Specifically, we examine whether retail investors pay more attention to stocks with (stronger) salient payoffs and explore their trading behavior of different beta stocks in the presence of salient upsides and downsides. Following Da, Engelberg and Gao (2011), we use abnormal Google search volume to proxy retail investor attention. We find that the retail attention is significantly higher among stocks with salient upsides and downsides than non-salient stocks, especially for stocks with salient upsides. For stocks with both salient upsides and downside, the attention generally increases with beta, with the high beta stocks being received the most attention. Following Boehmer, Jones, Zhang and Zhang (2021), we use scaled retail buy-sell order imbalance to measure the trading behavior of retail investors. We find supportive evidence that retail investors demand more high beta stocks than low beta stocks in the presence of salient upsides, contributing to stronger overpricing of high beta stocks. However, they demand less high beta stocks in the presence of salient downsides.

We next investigate a range of alternative explanations that could potentially account for our empirical finding. First, we investigate whether our salience measure indirectly reflects investor preference for lottery-like stocks. Bali, Brown, Murray, and Tang (2017) find that lottery preference drives stocks with high beta to earn lower expected returns than those with low beta. This is because high beta stocks generally tend to have higher skewness. The lower return of high beta stocks is mechanically driven by the overdemand from investors with lottery preference. Since our salience measure is positively correlated with the lottery preference proxies, our empirical finding could be partly driven by the lottery preference mechanism.

The second explanation we consider is the salience induced mispricing and limits to arbitrage. Due to the salience effect, investors are attracted to stocks with salient upsides, which are subsequently overpriced, while they stay away from stocks with salient downsides, resulting in underpricing. High beta stocks tend to be associated with higher arbitrage costs and thus are more susceptible to mispricing compared to low beta stocks. Taken together, among stocks with salient upsides, high beta stocks would be more overpriced than low beta stocks, leading to a negative beta-return relation while among stocks with salient downsides, high beta stocks would be more underpriced than low beta stocks, resulting in a positive beta-return relation. Thus, the salience induced mispricing and limits to arbitrage would potentially capture our empirical pattern. We posit that high-risk firms inherently tend to produce salient returns, which is different from the notion of salience induced mispricing and limits to arbitrage. Hence, risk itself does not necessarily act as a proxy for arbitrage costs that exacerbate the mispricing stemming from salience.

Third, Wang, Yan and Yu (2017) show that the reference-dependent risk preference is an important determinant of the risk-return relation. This proposition is based on prospect theory and mental accounting that investors exhibit risk-seeking behavior when they face capital losses and risk-averse behavior when they face capital gains. The mental accounting provides a reference for investors to set different accounts to determine gains and losses. They show that the risk-return relation is positive (negative) among stocks with capital gains (losses).

At last, we consider other two prominent explanations for the beta anomaly from the previous literature, coskewness risk and disagreement effect. Schneider, Wagner and Zechner (2020) contend that the CAPM's failure stems from its neglect of compensating for negatively priced coskewness risk. Hong and Sraer (2016) show that in the presence of short-sale constraints and disagreement about aggregate market prospects, high beta stocks are more speculative and more likely to be overvalued. They also show the negative security market line due to the disagreement induced overpricing appears only among speculative stocks.

To examine whether our empirical patterns are driven by those possible mechanisms, we adopt two approaches. The first approach is to perform a battery of Fama-MacBeth regressions by taking account of a wide range of proxies for lottery preference, reference-dependence preference (RDP), limits to arbitrage, coskewness risk and stock speculativeness. We find the salience effect on the beta-return relation remains highly significant. The second approach is to use a residual salience measure controlling for other potential mechanisms simultaneously by performing a cross-sectional regression on those variables. We then double sort on beta and the residual salience and find the economic magnitude and significance of the results are essentially similar to before. In sum, our empirical patterns are not likely driven by other possible mechanisms, and instead the salience effect is a key determinant of the beta-return relation.

Furthermore, outside of the beta-return framework, we uncover widespread evidence of this behavioral bias in other risk-return relationships, where risk is gauged using total volatility and uncertainty risk proxies (Zhang, 2006; Wang, Yan, and Yu, 2017).

Our major contribution to the literature is that we empirically reconcile the insignificant or negative beta-return puzzle. Many studies try to explain the puzzling beta-return relation including lottery preference (e.g., Bali et al., 2017), disagreement (e.g., Hong and Sraer, 2016), investor sentiment (e.g., Yu and Yuan, 2011), inflation (e.g., Cohen, Polk, and Vuolteenaho, 2005), leverage constraints (e.g., Frazzini and Pedersen, 2014), arbitrage asymmetry (e.g., Stambaugh, Yu and Yuan, 2015) and coskewness risk (e.g., Schneider, Wagner and Zechner, 2020). We propose that the salience effect is a key determinant for the risk-return trade-off and does not fall in either of these categories of above explanations. We complement the previous literature by showing that the salience effect is one of the most economically important drivers of risk anomalies. However, unlike explanations in the previous literature, we show that high-risk firms can be either overpriced or underpriced, depending on whether their high or low returns are salient.

Our work adds to the growing literature on the role of salience effect in decision-making and asset pricing. BGS (2012, 2013) develop the salience theory by showing investors engage in salient thinking and make decisions in a context. They show that the salience model can theoretically account for many empirical phenomena, such as the Allais paradox, preference reversals, skewness preference and time-varying risk premia. BGS (2021) review the psychology of bottom up attention and regard salience as the property of a stimulus that draws attention bottom up. Bottom up shifts of attention generate systematic instability in risk attitudes. Cosemans and Frehen (2021) provide the first empirical evidence of salience theory by showing that high salient firms significantly underperform their low salient counterparts.

Cakici and Zaremba (2021) find supportive evidence of salience theory in international markets, but they find that the salient anomaly can be attributed to the short-term return reversal, it is priced primarily among microcaps, the premium is realized predominantly following severe down markets and volatility spikes. Outside of microcaps and extreme market conditions, the salience phenomena do not exist. Distinct to Cosemans and Frehen (2021) and Cakici and Zaremba (2021) where they focus on the predictability of the salience measure to the cross-section stock returns, we focus on the salience effect on the risk-return relationship. In our later analysis of the salience effect on the risk-return relation, we take firm size, short-term reversal, extreme market states and other issues mentioned in Cakici and Zaremba (2021) into consideration and find our results remain robust after taking account of those issues.

Lastly, our paper is related to the extensive literature of cumulative prospect theory (Tversky and Kahneman, 1992) regarding the overweighting of small probabilities by decision markers. Barberis and Huang (2008) model that cumulative prospect theory investors overweight small probabilities of extreme high returns, inducing a strong preference for lottery-like stocks. This causes those stocks to be overvalued and to earn subsequent lower returns. Barberis, Mukherjee and Wang (2016) find empirical evidence supporting this notion. Bali et al. (2017) show that the overpricing of high beta stocks is driven by lottery preference. However, our salience theory in explaining the beta-return relation is distinct from the lottery preference due to the small probability weighting from prospect theory in two aspects. First, in contrast to prospect theory, low probability payoffs are not always overweighted in the salience model; they are overweighted only if they are salient, regardless of probability. That is, overweighting depends not only on the probability of a state but also on the salience of its payoff. Consequently, the lottery upside may still be underweighted if the payoff associated with it is not sufficiently high to be salient. Second, more importantly, investor's decision weights are influenced by the context in which they are presented with available alternatives. We show that the salience effect in determining the risk-return trade-off is not affected by the lottery preference in the regression analysis and using the residual salience measure controlling for lottery proxies.

The rest of the paper is organized as follows. Section 2 discusses the salience theory and develops our hypothesis. Section 3 describes the data and salience theory measure. Section 4 conducts our main empirical analysis and provides evidence on the salience effect on the beta-return relation. Section 5 examines a range of alternative explanations for our main empirical finding. Section 6 demonstrates that the same behavioral bias also applies to other risk-return settings. Section 7 concludes.

2. Salience theory and hypothesis development

We discuss salience theory and motivations of how salience behavior affects the risk-return relation. The salience theory of BGS (2012, 2013) has recently attracted substantial attention in the literature and accounts for many puzzles[2]. The theory argues that risky assets are evaluated in a context described as available alternatives in the market. In comparisons between risky assets, individuals focus their attention to the payoffs that are most different or stand out relative to the available alternatives in the market. A stock's salient payoff is therefore the one that is different from the average payoff in a given state of the world. Intuitively, the more it stands out from other stock returns in the market, the stronger influence on the subjective decision weights placed by investor will be. The idea is motivated by Kahneman (2003) who

[2] For example, those puzzles include Allais paradox, skewness preference, value anomaly and time-varying risk premia.

argues "changes and differences are more accessible to a decision maker than absolute values". In making choices, investors overweight these salient payoffs relative to their objective probabilities and neglect the non-salient payoffs.

In the salience model, risk attitudes are context-dependent which are driven by the salience of different stock payoffs. When stocks' upside returns are salient, investors are attracted to these stocks and exhibit risk-seeking behavior. The excess demand for these stocks results in overvaluation and lower future returns. In contrast, when stocks' downside returns are salient, investors are risk-averse and demand a positive risk premium for holding those stocks. Please see Appendix for a demonstration of incorporating salience into a standard asset pricing model.

We hypothesize that investors' risk attitudes and resulting demand for high-risk versus low-risk stocks are driven by the salience of those stocks' upside or downside returns. Specifically, among stocks exhibiting salient upside returns, investors will exhibit risk-seeking behavior according to salience theory. They will overweight the salient upside payoffs, which are likely to be larger in magnitude for high-risk stocks. This is because high-risk stocks, by nature of their higher volatility and wider distribution of returns, are more prone to experiencing extreme positive returns that stand out from the average or norm. Consequently, investors will be more attracted to holding these high-risk stocks exhibiting salient upside returns, compared to their attraction towards low-risk stocks. This risk-seeking behavior will drive higher demand and overvaluation of the high-risk stocks initially, leading to lower subsequent returns over time, resulting in a negative risk-return relationship.

On the other hand, among stocks with salient downside returns, investors will exhibit risk-averse behavior. They will overweight the salient downsides, which again are likely to be larger for high-risk stocks due to their inherent volatility. High-risk stocks are more likely to experience extreme negative returns that deviate significantly from the average or norm. As a result, investors will be relatively less attracted to holding these high-risk stocks exhibiting salient downside returns, compared to low-risk stocks. This risk-averse behavior will lead to lower demand and undervaluation of the high-risk stocks. However, this undervaluation will correct over time, resulting in a positive risk-return relationship.

We further expect that the negative risk-return relation for stocks with salient upsides tends to more pronounced than the positive risk-return relation for stocks with salient downsides. This is because of short-sale constraints, making it more costly for arbitrageurs to exploit overpricing of stocks with salient upsides. However, arbitraging underpricing of stocks with salient downsides only requires buying. Taken together, we propose our main hypothesis as follows:

Hypothesis: *The risk-return relation should be negative among stocks with salient upsides but positive among stocks with salient downsides. Due to short-sale constraints, the negative relation is more pronounced than the positive one.*

3. Data and Methodology

3.1. Stock market data

The data are collected from several data sources. Our sample includes all common stocks (CRSP share codes 10 and 11) listed on the NYSE, AMEX, and NASDAQ (CRSP exchange codes 1, 2, and 3) from the Centre for Research in Security Prices (CRSP) daily and monthly files. The sample period spans July 1963 to December 2018. We exclude financial firms (SIC codes between 6000 and 6999) and firms with negative book equity. To reduce the potential effects of penny stocks, we keep only stocks with a price of at least $1 per share. Stock return data are from CRSP, and accounting data are from Compustat. Institutional ownership data are from the Thomason Financial 13F file from January 1980 to December 2018. The stock's market beta (BETA) at the end of each month is estimated as the slope coefficient of a regression of monthly excess stock returns on market excess return over a 60-month rolling window with a minimum of 24 months.[3]

3.2. Salience theory measure

In constructing the salience theory measure, we follow Cosemans and Frehen (2021). As future payoff distributions are not directly observable, past one-month return distribution is employed as a proxy for its future return distribution.[4] The state space is proxied by daily returns over the past month, with the objective probability for each state being the reciprocal of the number of trading days in the month.

Suppose there is a stock *i*. The salience of its payoff depends on how its return is different to the average market payoff with all available stocks in the same state. We use the return of equal-weighted CRSP index which includes all stocks in the CRSP universe as the average market payoff.[5] The salience function proposed by BGS (2012) is as follows:

$$\sigma(r_{i,s}, r_{m,s}) = \frac{|r_{i,s} - r_{m,s}|}{|r_{i,s}| + |r_{m,s}| + \theta} \quad (1)$$

where $\theta > 0$, $r_{i,s}$ is the stock *i*'s return on day *s* and $r_{m,s}$ is the market return of equal-

[3] Alternative beta measures are estimated in Section 4.5, and our empirical findings remain robust.

[4] We construct alternative ST measures to test its robustness in Section 4.5. Our conclusion remains the same.

[5] The results are robust using value-weighted CRSP index.

weighted CRSP index on day s. This equation implies that the salience is based on an individual stock payoff relative to the market payoff. The salience function of Eq. (1) satisfies three properties: (1) ordering: the salience of state s of stock i increases in the distance between its payoff $r_{i,s}$ and the market payoff $r_{m,s}$. This is captured by the numerator $|r_{i,s} - r_{m,s}|$; (2) diminishing sensitivity: the salience decreases as the average market payoffs gets further from zero in absolute value as captured by the denominator $|r_{i,s}| + |r_{m,s}|$; (3) reflection: the salience is determined by the magnitude rather than the sign of payoffs. Both significantly different gains and losses can cause stock payoff to be salient. The payoff $r_{i,s1}$ that stock i pays in state s_1 is more salient than the payoff $r_{i,s2}$ in state s_2 if and only if $\sigma(r_{i,s1}, r_{s1}) > \sigma(r_{i,s2}, r_{s2})$.

In the valuation of stock i by investors who are subject to salient-thinking behavior, they rank the states of stock i according to the salience function of each state (Eq. (1)) from 1 (most salient) to n (least salient), with n denoting the number of trading days in the month. The ranking is denoted as $k_{i,s}$. Then we use the ranking to calculate the salience distorted weight of stock i. Salient thinkers put more weights on stock's salient payoff at a particular state rather than its objective state probability as follows:

$$\hat{\pi}_{i,s} = \pi_s * \omega_{i,s} \quad (2)$$

where $\hat{\pi}_{i,s}$ denotes as distorted salience probability by the salience thinkers, π_s is the objective probability and $\omega_{i,s}$ is the salience weight defined as

$$\omega_{i,s} = \frac{\delta^{k_{i,s}}}{\sum_{s'} \pi_{s'} \delta^{k_{i,s'}}} \quad (3)$$

where $\delta < 1$ which captures the degree to which the salient thinker draws attention to salient payoffs and neglects non-salient payoffs. If $\delta = 1$, the weight $\omega_{i,s}$ is equal to 1 for all states s, implying there is no salience distortions. This is the case of a rational investor. If instead δ

< 1, the salient thinker overweights salient payoffs and underweights non-salient payoffs. In the case of $\delta \to 0$, the salient thinker only takes account of a stock's most salient payoff and neglects all other payoffs. We follow BGS (2012) to specify the values of θ and δ of 0.1 and 0.7, respectively.[6]

At last, the salience theory value (ST) for each stock in month t by calculating the covariance between salience weights and daily returns as follows:

$$ST_{i,t} = cov[\omega_{i,s,t}, r_{i,s,t}] = \sum_{s \in S_t} \pi_{s,t} \omega_{i,s,t} r_{i,s,t} - \sum_{s \in S_t} \pi_{s,t} r_{i,s,t} \quad (4)$$

where the first term captures salience expected return of month t computed by salience-weighted past daily returns and the second term is expected return computed by equal-weighted past daily returns. The ST captures the salience distortion in return expectations driven by salience behavior. When the stock return is salient upside, investors exhibit risk-seeking behavior whereas when its return is salient downside, investors are risk-averse.

3.3. Retail investor attention

To measure retail investor attention, we follow Da, Engelberg and Gao (2011) and use Google search volume (SVI) for each stock. The SVI is a relative search popularity score scaled between 0 to 100. Da, Engelberg and Gao (2011) argue that retail investors tend to use Google to gather information. We use CRSP stock tickers instead of firm names and exclude all tickers which are classified as English words to ensure that the search results are indeed for the stocks and not for other generic items. We measure monthly abnormal retail attention (ASVI) for a stock as the log of SVI during the month minus the log of median SVI during the previous six

[6] Our results are robust to alternative choices of θ and δ.

months. To make sure our results are not driven by extreme values of ASVI, we winsorize ASVI at the 1% level. The data of ASVI is available from 2004.

3.4. Retail investor trading activity

We follow Boehmer, Jones, Zhang and Zhang (2021) to measure the trading behavior of retail investors. The measure is developed considering regulatory restrictions in the U.S (Reg NMS) and the resulting institutional arrangements, whereby retail order flows, but not institutional order flows, can receive sub-penny price improvements. Following Boehmer et al. (2021), we use the trade data from FINRA TRF and estimate the fraction of the penny associated with the transaction price (P_{it}): $Z_{it} \equiv 100 * \text{mod}(P_{it}, 0.01)$. If Z_{it} is in the range of (0, 0.4), the trades are classified as retail sell transactions. If Z_{it} is in the range of (0.6, 1), the trades are classified as retail buy transactions. The rest of trades with Z_{it} is in the range of (0.4, 0.6) do not belong to the retail investor trading category.

We calculate the daily retail buy-sell order imbalance for a stock as the difference between the number of buy trades minus the number of sell trades. The difference is scaled by the number of shares outstanding as reported by CRSP to make sure the measure to be directly comparable across stocks (e.g., McLean, Pontiff and Reilly, 2020). We then aggregate this daily measure to the monthly measure by taking average of the daily measure within that month and denote the monthly buy-sell order imbalance as RBSI. At last, we winsorize RBSI at the 1% level to reduce the impact of potential outliers. The data is available from 2010.

3.5. Summary statistics and single sort

Panels A and B of Table 1 present summary statistics of main variables in this study. ST has a mean of 0.7% and a standard deviation of 3.1%. The average BETA is almost 1, and ST and BETA is positively but weakly correlated. This largely alleviates the concern that sorting by ST is potentially a further sort by BETA in Section 4.1 double-sort analysis. In addition, ST is positively correlated with idiosyncratic volatility (IVOL) and idiosyncratic skewness (ISKEW) which are skewness proxies in the literature (e.g., Kumar, 2009). Section 5.2 shows our mechanism is distinct from the lottery/skewness preference in affecting the beta-return relation.

[Insert Table 1 about here]

Panel C of Table 1 reports the results of portfolios sorted on BETA. We sort stocks into quintiles based on their BETA. We hold portfolios for one month and equal weight portfolio returns. Consistent with previous studies, there is no significant or negative relation between beta and return depending on benchmarks. The difference between high and low beta portfolios is -0.442% (t = -2.86) and -0.048% (t = -0.34) with respect to Fama-French 3- and 5-factor risk adjustments, respectively.

4. Empirical analysis

In this section, we first report monthly excess returns and Fama-French 5-factor (FF-5) alphas for portfolios formed based on beta and salience theory measures. This analysis helps us to determine the economic magnitude of the role of salience effect in determining the beta-return relation. We then examine in a cross-sectional regression framework that controls for a range of firm characteristics. Next, we provide direct evidence by examining retail investor attention and trading behavior. Finally, we show that the role of salience on the beta-return relation survives a number of robustness checks.

4.1. Double sorts

To examine the importance of salience behavior on the beta-return relation, we begin with the portfolio analysis. We double sort stocks sequentially into quintiles first based on BETA and then further divide each beta quintile into ST sub-portfolios.[7] The dependent sorts result in 25 beta-salience portfolios.[8] We hold portfolios for one month and both equal- and value-weight portfolio returns. We expect to see the beta-return relation is negative for stocks with salient upsides and positive for stocks with salient downsides.

Table 2, which contains our key results, reports average ST, excess return and FF-5 alpha for each of the 25 portfolios. The empirical results provide strong support to our hypothesis. Panel A of Table 2 reports the average stock ST value within each portfolio. We observe that high beta stocks tend to have significantly stronger salient upside (high ST) and downside (low ST) returns than low beta stocks, which is consistent with our hypothesis.

Panels B and C of Table 2 report the excess return and FF-5 alpha of equal-weighted portfolios. The results are similar but even sharper after adjusting FF-5 risk factors. The rest of results for portfolio analysis are presented using FF-5 alphas unless stated. We can see from Panel C of Table 2 that for stocks with salient upsides (high ST quintile), the FF-5 alphas are monotonically decreasing in beta, with the difference between high and low beta quintiles equals to -0.778% (*t*-statistic = -4.13). For stocks with salient downsides, the average alphas are generally increasing in beta, with the difference between high and low beta portfolios equals to 0.465% per month (*t*-statistic = 2.65). For stocks in the middle salience quintile, there is no

[7] As a robustness check, we use a sequential sort first on ST then BETA. The results are reported in Table A1 and remain similar.

[8] The results are qualitatively similar by forming 3×3 portfolios based on the dependent sorts of BETA and ST.

apparent beta-return pattern, with the difference between the two extreme beta portfolios being only 0.174 (*t*-statistic = 1.27). The role of salience effect in determining the direction and magnitude of beta-return relation is shown in Figure 1, which depicts the FF-5 alphas of beta portfolios across the salience quintiles.

[Insert Table 2 about here] & [Insert Figure 1 about here]

Moreover, evident in Table 2 and Figure 1 is the asymmetric strength of the beta-return relation across salience portfolios. Due to short-sale constraints, the negative beta-return relation tends to be stronger than the positive relation. Given the asymmetric strength of the negative and positive beta-return relations among salient upside and downside stocks, aggregating across all stocks leads to a slightly negative beta-return relation shown in Panel C of Table 1.

Many studies document possible unconditional overpricing mechanisms that drive the insignificant or negative beta-return relation. For example, Bali et al. (2017) show that the irrational demand for high beta stocks is driven by investor preference for lottery-like stocks, leading to inflated price for high beta stocks and subsequently lower future returns. Moreover, high risk stocks are more sensitive to investor disagreement and sentiment (e.g., Hong and Sraer, 2016; Shen, Yu, and Zhao, 2017). Due to short-sale constraints, those stocks are overpriced. Unlikely those studies in the literature, we show that high-beta stocks can be both overvalued and undervalued. They tend to be more overvalued than low-beta stocks when their high returns are salient but to be more undervalued than low-beta stocks when their low returns are salient. This is primarily attributed to their volatile returns, which make their high and low returns more salient, consequently leading to more pronounced mispricing.

The sequential double sorts in Panels B and C of Table 2 investigate the marginal impact of salience on the beta-return relation. There may be a possibility that the second sort on ST may produce further variations in beta that explains the difference in the beta-return relation among stocks with salient upsides and downsides. Thus, as a robustness check, Panel D of Table 2 reports the alphas using independent double sorts on BETA and ST. The independent double sorts in Panel D generate similar empirical patterns, with the beta-return relation being significantly positive for stocks with salient downsides and negative for stocks with salient upsides.

To ensure our results are not driven by small stocks, we first value-weight portfolio returns, and the results are shown in Panel E of Table 2. The beta-return relations are still positive and negative among stocks with salient downsides and upsides respectively. As expected, the results are less pronounced than equal-weight portfolios because large stocks tend to be less subject to the salience behavioral bias by individuals and limits to arbitrage. Next, we delete bottom 20% stocks based on their market capitalization and the results are reported in Table A2. Again, the beta-return relation remains negative (positive) among stocks with salient upsides (downsides).

4.2. Fama-MacBeth regression

The double-sort results in the previous section are indicative of the salience effect on the beta-return relation, but it cannot explicitly control for other variables that may potentially affect returns. In this section, we conduct a series of Fama and MacBeth (1973) regression analyses. In particular, we estimate predictive regressions using the following specification:

$$\mathrm{r}_{i,t+1} = \alpha + \beta_1 \mathrm{BETA}_{i,t} + \beta_2 \mathrm{ST\_HIGH}_{i,t} + \beta_3 (\mathrm{BETA} \times \mathrm{ST})_{i,t} + \beta_4 \mathrm{SIZE}_{i,t}$$
$$+ \beta_5 \mathrm{BM}_{i,t} + \beta_6 \mathrm{RET1}_{i,t} + \beta_7 \mathrm{RET12}_{i,t} + \beta_8 \mathrm{RET36}_{i,t} + \beta_9 \mathrm{ILLIQ}_{i,t}$$
$$+ \beta_{10} \mathrm{TURN}_{i,t} + \epsilon_{i,t} \qquad (5)$$

where $\mathrm{r}_{i,t+1}$ is stock return of stock *i* in month *t*+1, $\mathrm{BETA}_{i,t}$ is the CAPM beta of stock *i* estimated in month *t*, $\mathrm{ST}_{i,t}$ is the salience theory measure of stock *i* in month *t*, $\mathrm{SIZE}_{i,t}$ is the natural log of market capitalization at the end of month *t*, $\mathrm{BM}_{i,t}$ is the natural log of book-to-market ratio of stock *i* in month *t*, $\mathrm{RET1}_{i,t}$ is return of stock *i* in month *t*, $\mathrm{RET12}_{i,t}$ is cumulative returns of stock *i* from months *t-12* to *t-1* by skipping month *t*, $\mathrm{RET36}_{i,t}$ is cumulative returns of stock *i* from months *t-13* to *t-36*, $\mathrm{ILLIQ}_{i,t}$ is the Amihud's (2002) illiquidity of stock *i* in month *t*, $\mathrm{TURN}_{i,t}$ is turnover of stock *i* in month *t*. The construction of those variables is discussed in detail in the appendix. Following Conrad, Kapadia, and Xing (2014) and An, Wang, Wang and Yu (2019), the independent variables are winsorized at 5% and 95% to avoid extreme values of the variables affecting our results.[9] The *t*-statistics are calculated based on Newey and West (1987).

[Insert Table 3 about here]

Table 3 reports the regression results. The variable of interest is the interaction between BETA and ST. As it is hypothesized that the beta-return relation should be negative (positive) for stocks with salient upsides (downsides), we expect the interaction term to be negative. Moreover, we replace continuous variable of ST with two ST dummies, salient upsides (ST_HIGH) and salient downsides (ST_LOW). We expect the interaction between BETA and

[9] The results based on an alternative winsorization at 1% and 99% are essentially similar. In Section 4.5, we show that our results are not affected by winsorization.

ST_HIGH to be negative and that between BETA and ST_LOW to be positive.

The first two regression models report the results by excluding the interaction term to examine whether BETA and ST are individually associated with future returns. The coefficient of BETA is insignificant, confirming with the insignificant beta-return relation in the literature. The ST coefficient is negative and significant at the 1% level, consistent with Coseman and Frehen (2021). Models (3) and (4) include the interaction terms. In model (3), the interaction between ST_HIGH and BETA (ST_LOW and BETA) is significantly negative (positive), suggesting the beta-return relation is significant negative (positive) among stocks with salient upsides (downsides). In model (4), the interaction term between BETA and ST is highly significant, with the *t*-statistic being larger than the threshold of 3 proposed by Harvey, Liu, and Zhu (2016). A one standard deviation decrease in ST adds 0.35% to the predictive power of BETA on a monthly basis after controlling for major firm characteristics. This figure accounts almost 6 times to the predictive value of BETA by itself. In sum, both our portfolio and regression results show the salience effect is important in determining the beta-return relation.

4.3. Evidence from market states

In the previous section, we find the beta-return relation crucially depends on the salience of stock returns. Recall our hypothesis that high-risk firms, as opposed to low-risk firms, are likely to experience both salient upsides and downsides in their returns, and those returns tends to be more salient in magnitude. Therefore, high-risk firms tend to have stronger mispricing, i.e., underpricing in the presence of salient downside returns and overpricing in the presence of salient upside returns according to BGS (2012).

Here we examine the salience effect under different market states in terms of high and low market volatility. We expect the salience effect on the beta-return relation to be stronger in highly volatile market states compared to less volatile market states. This is because the stock's salient return is more likely and/or tends to be stronger during high market volatility states. To do so, we first classify sample periods into high and low market volatilities based on the median of volatilities, which are estimated using the 6-month value-weighted market returns over the entire sample period. We then perform the double-sort analysis as we did in Panel C of Table 2, in both high and low market volatility states.

[Insert Table 4 about here]

The results are reported in Table 4 and are consistent with our conjecture. Among stocks with salient upsides, the alpha spread between high and low beta portfolios is -0.813% in high market volatility compared to -0.771% in low market volatility states. Among stocks with salient downsides, the positive beta-return relation is stronger in high market volatility than in low market volatility. Specifically, the alpha spread between high and low beta alphas is significant 1.016% in high volatile states compared to insignificant -0.046% in low volatile states – a difference of 1.062%, with a *t*-statistic of 3.23. The beta spread between high and low ST is significantly higher in high market volatility than low market volatility (-1.830% vs -0.725%) states.

Cakici and Zaremba (2021) find that the salience effect concentrates primarily on volatile periods. Our results in Table 4 show that while the salience effect on the beta-return relation is much stronger in high market volatility states than low volatility states, the effect of salience on the beta-return relation exists in both market states.

4.4. *Evidence from retail investor attention and trading*

BGS (2012) argue that investors tend to engage in salient thinking and draw their most attention to the stock salient upside and downside payoffs, and their risk attitudes are driven by the salience of stocks payoffs. As retails investors are more prone to psychological bias (e.g., salient thinking behavior), we focus on the behavior of retail investors. Specifically, we provide direct evidence that retail investors pay more attention to stocks with salient payoffs and examine their trading behavior for different beta stocks in the presence of salient upsides and downsides.

[Insert Table 5 about here]

First, the findings in panel A of Table 5 confirm that retail investors indeed pay considerably more attention for stocks with both salient upsides and downsides. We double sort stocks into quintiles based on BETA and ST in month t and calculate equal-weight retail investor attention in month t as proxied by abnormal Google search volume (ASVI). The retail attention is significantly higher among stocks with salient upsides and downside than non-salient stocks, especially for stocks with salient upsides. For stocks with salient downsides, the retail attention generally increases with beta. For stocks with salient upsides, the attention is monotonically increase across beta portfolios, with the high beta stocks received the most attention.

Next, we investigate how retail investors trade high and low beta stocks in the presence of salient upsides and downsides. Based on our hypothesis that high beta stocks are more likely to yield both salient upside and downside returns than low beta stocks, we expect that retail investors demand more high beta stocks than low beta stocks in the presence of salient upsides but demand less high beta stocks in the presence of salient downsides. In Panel B of Table 5, we perform the Fama-MacBeth regression by regressing of scaled retail buy-sell order imbalance (RBSI) in month *t* on BETA, ST and their interaction term in month *t*. We also control for other firm characteristics since investor demand maybe driven by those characteristics. The results are reported in Panel B of Table 5. Column (1) reports the results using the continuous ST variable. Consistent with our conjecture, we find that the coefficient of interaction term of BETA and ST is significantly positive, and that of BETA is significantly negative, suggesting that when salience is high, investors tend to demand more high beta stocks than low beta stocks and when salience is low, investor instead demand less high beta stocks.

To see how retail investors demand for high and low beta stocks within each category of salience payoffs, we create a category variable of ST (ST_C) which equals 0 to 4 with 0 being the category of stocks with salient downsides and 4 being the category of stocks with salient upsides. The results using the ST_C in column (2) shows the similar pattern to those in column (1) using ST. When stock return is salient upside (e.g., ST_C = 4), the demand for high beta stocks is significantly higher than low beta stocks. In contrast, for salient downsides (e.g., ST_C = 0), the demand for high beta stocks is indeed significantly lower as the coefficient of BETA is significantly at -0.027. The finding of retail investor demand provides strong support to our hypothesis that high-risk stocks tend to be more overvalued and undervalued in the presence of salient upsides and downsides because of their volatile returns.

Our finding is related to Barber and Odean (2008) who argue that individual investors are reluctant or unable to short sell and are net buyers of attention-grabbing stocks, resulting in positive price pressure in the short run and subsequent reversal. Based on their attention theory, investors would demand similar amounts of both high and low beta stocks in the presence of salient downsides as retail investor attention is similar for both high and low beta stocks, which is inconsistent with our findings in the Panel B of Table 5. Instead, we show that while retail investors pay significant much more attention for stocks with salient upsides and downsides (in line with the theory of BGS (2012)), they actually exhibit distinct trading behavior for high and low beta stocks across different salient payoffs.

4.5.Robustness checks

In this section, we perform a series of additional tests to assess the robustness of our results about the salience effect on the beta-return relation. For brevity, we only report the coefficient of our main variable of interest, which is the coefficient of the interaction term of BETA and ST from the Fama-MacBeth regression as in model (4) of Table 3.

First, many studies employ different approaches for estimating beta for individual stocks. To ensure our results are not sensitive to a specific beta estimation, we use three alternatives which are widely used in the literature in addressing the beta-return relation. The first alternative estimation is from Vasicek (1973). Specifically, we regress stock excess returns on market excess returns over a rolling window of 60 months with a minimum of 24 non-missing data to compute time-series beta ($\widehat{\beta_i^{TS}}$) as in our main analysis. Further, a Bayesian shrinkage factor is used to reduce the influence of outliers and is given by

$$\widehat{\beta_i} = w_i\widehat{\beta_i^{TS}} + (1 - w_i)\widehat{\beta_i^{XS}} \quad (6)$$

where $w_i = \frac{1-\sigma_{i,TS}^2}{\sigma_{i,TS}^2+\sigma_{XS}^2}$ and $\widehat{\beta_i^{XS}}$ is the cross-sectional average of time-series beta in each month. $\sigma_{i,TS}^2$ is the variance of the beta for each stock and σ_{XS}^2 is the variance of beta across stocks in each month. Such a beta estimation method is also used by Liu, Stambaugh and Yuan

(2018). The second method is to accommodate nonsynchronous trading effects (Dimson, 1979). Following Fama and French (1992), we estimate beta as the sum of slopes in the regression of the excess return of a stock on the current and past month market excess return over the most recent 60 months with a minimum of 24 non-missing data and then apply for Vasicek's (1973) shrinkage method. The third estimation method is according to Frazzini and Pedersen (thereafter, FP) (2014). For each stock, the beta is estimated as follows:

$$\widehat{\beta_i^{TS}} = \hat{\rho}\frac{\widehat{\sigma_i}}{\widehat{\sigma_m}} \qquad (7)$$

where $\widehat{\sigma_i}$ and $\widehat{\sigma_m}$ are volatilities of stock *i* and of the market excess returns, respectively, and $\hat{\rho}$ is their correlation. The volatilities are estimated as one-year rolling standard deviations of one-day log returns with a minimum of non-missing 120 observations and the correlation is estimated over a five-year rolling window using overlapping three-day log returns with a minimum of 750 non-missing observations. Following FP (2014), Vasicek's (1973) shrinkage method is applied with w=0.6 and β^{XS}=1.

[Insert Table 6 about here]

We perform the Fama-MacBeth regressions by controlling for other firm characteristics in the form of model (4) of Table 3 but replace BETA with alternative betas. The results are shown in the first three models of Table 6. In all cases, the interaction term is significantly negative at the 5% level or above. We then repeat the double-sort analysis in Table A3 using alternative betas. The beta-return relation is significantly negative among stocks with salient upsides in all three approaches and significantly positive among stocks with downsides in two cases (not significant using the FP's (2014) method). In sum, the salience effect on the beta-return relation is robust to alternative beta estimations.

Second, we analyze whether our results are sensitive to particular stock exchanges (i.e., NYSE/AMEX versus NASDAQ). The results appear in models (4) and (5) of Table 6. While the statistical significance for the interaction term is slightly lower due to a smaller number of observations, the salience effect remains strong and highly significant.

Third, we skip winsorization in model (6) and our results are not sensitive to the winsorization. Fourth, we exclude stocks with a share price less than $5 to further eliminate low price and small stocks in model (7). Our results are not affected by deleting those stocks. Moreover, to address microstructure issues in the regression estimations, we perform weighted least square (WLS) Fama-MacBeth regression by following Asparouhova, Bessembinder and Kalcheva (2010). Specifically, we estimate Eq. (5) by scaling observed returns by the gross return on the same stock in the previous month. Model (8) of Table 6 shows that the interaction between ST and BETA is still significant. Overall, we find our results continue to hold.

Fifth, previous literature (e.g., Bali, Cakici, Yan, and Zhang, 2005) documents that some market anomalies disappear once eliminating illiquid stocks. Therefore, to ensure our main findings are not driven by illiquid stocks, we exclude top decile illiquid stocks from the sample based on Amihud's (2002) illiquidity measure. Model (9) of Table 6 shows our empirical results continue to hold. Overall, our findings are not influenced by highly illiquid stocks.

Sixth, one potential concern is that beta is related to skewness and many studies (e.g., Kumar, 2009; Barberis and Huang, 2008; Bali, Cakici, and Whitelaw, 2011) show that investors have a particular preference for skewness/lottery stocks, resulting in overpricing of such stocks. BGS (2013) theoretically show that the salience which takes account of context evaluation can account of skewness preference. To ensure that our results are not mechanically driven by the positive association between skewness and beta[10], we compute residual BETA denoted RES_BETA in each month from a cross-sectional regression of BETA on idiosyncratic skewness and use RES_BETA in the regression analysis. The result is reported in model (10) of Table 6. The double-sort analysis is reported in Table A4. Our results using RES_BETA show that the beta-return pattern and its economic magnitude are essentially identical.

Seventh, models (11) and (12) report the results of a standard subperiod analysis. We divide the whole sample into two equal subperiods. The interaction terms are significant in both periods with the salience effect on the beta anomaly being stronger in the first period.

At last, we perform a number of analyses to assess the robustness of the salience measure. First, we compute ST using daily returns over longer horizons. In the baseline analysis, we assume investors form their future return expectation using daily returns over the past month since investors are more likely to recall the most recent month returns due to their cognitive limitations. Investors may also take account of more distant returns but believe them to be less representative of future outcomes. Using longer horizons in estimation also alleviates the concern that ST is driven by the short-term reversal effect. In models (13), (14) and (15), we use ST estimated over 3-, 6- and 12-month windows in the regression analysis. The interaction terms between ST and BETA remain highly significant in all cases.

We then examine whether the salience effect on the beta anomaly still exits when estimating ST using different frequencies. In models (16) and (17), we measure ST using weekly data over the last month and monthly data over the last 12-month, respectively. In both models, we find that the interactions of BETA and ST are negative and highly significant.

[10] For example, Panel B of Table 1 shows beta and idiosyncratic skewness is positively correlated.

Moreover, to alleviate the concern of bid-ask bounce, we drop the last day's return in calculating ST by following Cakici and Zaremba (2021). The results shown in model (18) are almost unaffected.

Overall, the salience effect on the beta-return relation is robust to different stock exchanges, skipping winsorization, exclusion of low price and highly illiquid stocks, WLS regression, the residual beta measure controlling for skewness, different subperiods as well as a range of alternative ST measures.

5. Alternative explanations

In this section, we consider several alternative explanations that could potentially affect our empirical finding documented in Section 4, including the lottery preference, limits to arbitrage, the role of reference-dependent preference, coskewness risk and disagreement effect.

5.1. Lottery preference

In this subsection, we explore the lottery preference explanation. Many studies show that individuals have a particular preference for lottery-like stocks (e.g., Kumar, 2009; Boyer, Mitton, and Vorkink, 2010). Barberis and Huang (2008), Bali, Cakici, and Whitelaw (2011) and Bali, Brown, Murray, and Tang (2017) show that the lottery preference can potentially account for stocks with high beta that tend to earn subsequent lower returns. The intuition is that high beta stocks are likely to be associated with higher skewness (i.e., lottery-like stocks). The irrational demand for high beta stocks is driven by investor preference for lottery-like stocks, pushing the price of high beta stocks higher and resulting in lower future returns.

Because our salience measure is positively correlated with lottery measures, a possible concern of our finding is that the role of salience might capture investors' lottery preference that is responsible for the negative beta-return relation. Empirically, we address this issue by adding lottery proxies and their interactions with BETA to Eq. (5).

We first consider three lottery measures: maximum daily return (MAX) following Bali, Cakici, and Whitelaw (2011), idiosyncratic skewness (ISKEW) and expected idiosyncratic skewness (ESKEW) following Boyer, Mitton, and Vorkink (2010). Table 7 reports the results. We have several observations from the table. First, without controlling for the salience effect, the interactions between lottery proxies (MAX and ISKEW) and BETA are both negative and highly significant, consistent with findings from previous studies. Second, the interaction between BETA and ST remains negative and highly significant even after controlling for all three lottery proxies and their interactions with BETA. Third, interestingly, after controlling for the interaction between BETA and ST, the interaction term between ISKEW and BETA is no longer significant and even reverses its sign. This indicates that the role of lottery preference may be partly driven by the salience effect on the beta-return relation. In sum, our results are robust after taking account of the role of lottery preference.

[Insert Table 7 about here]

5.2. The role of limits to arbitrage

Another explanation we consider is limits to arbitrage that prevent arbitrageurs to exploit the salience induced mispricing opportunities. Under the framework of BGS (2013), individuals are assumed to engage in salient thinking and draw most attention to the salient payoffs which stand out of the market. Consequently, they are attracted to stocks with salient upsides and stay away from stocks with salient downsides. Due to limits to arbitrage, this will lead to overpricing (underpricing) of stocks with salient upsides (downsides). Then rational investors will exploit such arbitrage opportunities by correcting mispricing, resulting in the underperformance (outperformance) stocks with salient upsides (downsides). When arbitrage is more limited, the salience induced mispricing would be stronger.

A possible concern of our findings is that our beta measure might indirectly reflect costs of arbitrage rather than the beta risk.[11] If this were to be the case, salience induced mispricing would be stronger among stocks with higher beta. Taken together, among stocks with salient upsides, high beta stocks are likely to be more overpriced than low beta stocks due to the role of limits of arbitrage, implying a negative beta-return relation. By contrast, among stocks with salient downsides, high beta stocks tend to be more underpriced than low beta stocks, implying a positive beta-return relation. This is potentially in line with the dual beta-return pattern observed in Section 4. Note that this mechanism does not rely on the fact of higher risk leading to stronger salient returns . It only requires the beta to be related to costs of arbitrage that prevent arbitrageurs from exploiting mispricing.

To address this concern, we control directly for the limits to arbitrage mechanism. We employ three limits to arbitrage measures. Our first measure is institutional ownership (IO) (see, e.g., Nagel, 2005; Berkman et al., 2009). The intuition for using IO is that institutional owners such as mutual funds and asset managers provide the main loan supply of stock, and thus stocks with low institutional ownership (i.e., fewer shares available for borrowing) are difficult or costly to short-sale (e.g., Chen, Hong, and Stein, 2002; Nagel, 2005; Hirshleifer, Teoh, and Yu, 2011). The second measure, illiquidity (ILLIQ), serves as a proxy for transaction

[11] For example, BETA is positively correlated with idiosyncratic volatility, a limits to arbitrage proxy.

costs (Amihud, 2002; Sadka and Scherbina, 2007). Intuitively, higher transaction costs dampen the profitability of arbitrage opportunities, which becomes less attractive to arbitrageurs. The third measure is idiosyncratic volatility (IVOL) which is widely used in the literature as a measure of limits to arbitrage (e.g., Pontiff, 2006). Moreover, Liu, Stambaugh and Yuan (2018) argue that the beta anomaly is primarily due to the positive correlation between IVOL and BETA. Since IVOL represents arbitrage risk, IVOL and return relation is negative among overpriced stocks but positive among underpriced stocks (Stambaugh, Yu and Yuan, 2015). The stronger overpricing combining with the positive BETA-IVOL produces the beta anomaly. The definitions and constructions of those three measures are provided in the appendix.

We rerun regression analysis by adding each of the three proxies and their interactions with ST to Eq. (5) to see whether the salience induced mispricing and limits of arbitrage impact the beta-return relation. Table 8 reports the results. Without controlling for the interaction between BETA and ST, the coefficient on the interaction term between the limits to arbitrage proxy and ST is found to be the expected sign and significant at the 10% level or above in all three cases. This suggests the salience induced mispricing is stronger among stocks with stronger limits to arbitrage. After controlling for those variables, our interaction term between BETA and ST remains negative and highly significant in all three specifications, suggesting that the role of limits of arbitrage does not account for our results.

[Insert Table 8 about here]

In sum, the salience effect in affecting the beta-return relation goes beyond the amplified salience induced mispricing by the role of limits to arbitrage.

5.3. *The role of reference-dependent preference*

Wang, Yan and Yu (2017) propose that the reference-dependent preference (RDP) accounts for the failure of risk-return relation. Prospect theory describes investors are risk-averse when they face capital gains and risk-seeking when they face capital losses. The mental accounting provides a reference for investors to set different accounts to determine gains and losses. In the domain of losses, due to limited arbitrage forces, high risk stocks could be overvalued by risk-seeking investors compared to low risk stocks, leading to lower future returns than low risk stocks. By contrast, in the domain of gains, the positive risk-return should hold as investors of these stocks are risk-averse. Therefore, RDP would lead to the same conclusion as the salience effect on the beta-return relation.

To make sure our salience effect is not driven by the RDP effect, we directly control for the RDP effect. Following Grinblatt and Han (2005), We construct the RDP effect proxy of capital gains overhang (CGO) to capture stocks' capital gains and losses. If the salience effect is driven by the RDP effect, then the interaction between BETA and ST should become insignificant after controlling for the interaction between BETA and CGO. Thus, we perform the Fama-MacBeth regression by adding CGO and its interaction with BETA to Eq. (5). Columns (1) and (2) in Table 9 report the results and indicate that controlling for the RDP effect has almost no impact on our ST interaction coefficient. Moreover, the coefficient of the interaction between BETA and CGO is significantly positive, consistent with Wang, Yan and Yu (2017). Overall, the role of salience in determining the beta-return relation is distinct from the RDP effect.

[Insert Table 9 about here]

5.4.Coskewness risk

Harvey and Siddique (2000) show that coskewness is negatively priced in the cross-section of equity returns, which implies that investors demand compensation for accepting negative skewness. Schneider, Wagner and Zechner (2020) find that the failure of the CAPM stems from its disregard of compensation for coskewness risk (COSKEW). After taking account of the coskewness risk, the beta-return relation becomes positive. In this subsection, we examine whether the salience effect on the beta-return relation is driven by coskewness risk. In accordance with Schneider, Wagner and Zechner (2020), COSKEW is measured as the slope coefficient from regression of the CAPM residual returns on squared market excess returns. We then perform the Fama-Macbeth regression analysis by adding COSKEW to Eq. (5). The results are reported in column (3) of Table 9. We find that the coefficient of interaction term is -11.291 (t = -5.35), which remains essentially unchanged. Thus, our finding is not influenced by the coskewness risk.

5.5. Disagreement effect

Hong and Sraer (2016) show that the negative relationship between beta and return is due to high sensitivity of high beta stocks to disagreement about the stock market's prospects and short-sale constraints. When disagreement is high, investors disagree more about the prospects of high-beta stocks compared to low-beta stocks, resulting in overpricing of high beta stocks due to short-sale constraints. Conversely, when disagreement is low, high beta stocks receive lower prices and higher expected returns because of risk sharing. They further show that the effect of disagreement on the beta-return relation only exist for speculative stocks. For nonspeculative stocks, the beta-return relation is not influenced by the disagreement effect.

To see whether our finding is affected the disagreement effect, we perform the Fama-MacBeth regression for both speculative and nonspeculative stocks. If the salience effect is not affected by the disagreement, we expect to observe similar beta-return patterns in both speculative and non-speculative subsamples. The speculativeness of a particular stock is defined as the ratio of stock beta to idiosyncratic variance following Hong and Sraer (2016). We first classify stocks into speculative and non-speculative stocks each month if their speculativeness is above and below the median respectively. We then perform the regression in Eq. (5) in both subsamples. The results are reported in columns (4) and (5) of Table 9. We find that the interaction between BETA and ST remains negative and highly significant in both subsamples. Therefore, our results of the salience effect on the beta-return relation are robust to the disagreement effect.

5.6. Residual salience theory measure

To ensure that our empirical findings in relation to ST are distinct from the existing mechanisms from the literature, we compute residual ST as RES_ST in each month from a cross-sectional regression of ST on a set of variables that have been shown associated with cross-sectional differences in the beta-return relation:

$$ST_{i,t} = \gamma_0 + \gamma_1 MAX_{i,t} + \gamma_2 ISKEW_{i,t} + \gamma_3 CGO_{i,t} + \gamma_4 COSKEW_{i,t} + \gamma_5 RET1_{i,t} + \varepsilon_{i,t} \qquad (8)$$

where for stock *i*, ST is the salience theory measure in month *t*, MAX is maximum daily return in month *t*, ISKEW is idiosyncratic volatility in month *t*, CGO is the capital gains overhang in month t, COSKEW is the coskewness risk.[12] Since our salience theory measure is constructed based on past one-month daily returns but with distorted weights and to ensure that the salience effect is not driven by the short-term reversal effect, we also take account of the short-term reversal effect in calculating RES_ST. We measure such an effect using the past-month return (RET1).

RES_ST is the residual ($\varepsilon_{i,t}$) from Eq. (8) for stock i in month t. It is intuitive that the residual salience is the measure controlling for the above mechanisms. Replacing ST with RES_ST, we repeat the portfolio analysis of Table 2 and still find a significantly positive beta-return relation among stocks with salient downsides and a negative relation among stocks with upsides. As shown in Panel B of Table 10, high beta stocks outperform low beta stocks by 0.334% (t-statistic of 1.88) among stocks with salient downsides and underperform low beta stocks by 0.613% (t-statistic of 2.87) among stocks with salient upsides. Those figures using RES_ST are similar to the corresponding figures albeit lower in magnitude using ST in Table 2. The evidence confirms that the salience effect in determining the beta-return relation is not driven by the potential mechanisms.

[Insert Table 10 about here]

6. Extensions: Salience and other risk-return relationships

In this section, we extend our analysis beyond the beta-return relation to other risk-return settings. We use several uncertainty proxies which are modelled as estimation of risk (e.g.,

[12] We don't include ESKEW in Eq. (8) in computing RES_ST as ESKEW has shown to have very little impact on the beta-return relation in Table 7. Our results remain essentially the same by including ESKEW in the calculation of RES_ST.

Barry and Brown, 1985; Coles and Loewenstein, 1988). Investors expect compensations for holding stocks with higher uncertainty risk (e.g., Knight, 1921; Anderson, Ghysels, and Juergens, 2009). However, the positive relationship is generally rejected by the data (e.g., Zhang, 2006). We include total volatility (TVOL), cash flow volatility (CVOL) and analyst coverage (COV) following Zhang (2006). Like beta and uncertainty risk, skewness is also used as an alternative risk measure. For example, Agarwal, Jiang and Wen (2021) find that underperforming fund increase their risk by investing in high skewness stocks. Following Boyer, Mitton, and Vorkink (2010), we employ expected skewness (ESKEW) to measure skewness risk. The definitions of alternative risk measures are described in the appendix. We also report the replication results of the original studies in Table A5 and find that there is no significant or negative relation between alternative risk proxies and return.

For each of the four risk-return relationships, we conduct a similar portfolio analysis as in Panel C of Table 2. Table 11 reports results of double-sorts under the four settings. Across all settings, we find that the relation between risk proxy and return is negative among stocks with salient upsides and positive among stocks with salient downsides. Specifically, for stocks with salient upsides, the difference between high and low risk portfolios based on TVOL, CVOL, COV and ESKEW and have monthly alphas of -1.199% (t = -6.22), -0.519% (t = -3.25), -0.8% (t = -4.25) and -0.881% (t = -2.91), respectively. In sharp contrast, when stocks are associated with salient downsides, high risk firms tend to earn higher alphas than low risk firms, with the spread between high and low risk firms being positive in all cases and significant in three out of the four cases. For example, high risk stocks outperform low risk stocks by 0.476% (t = 2.51) for TVOL, 0.237% (t =1.85) for CVOL, 0.259% (t = 1.45) for COV and 0.475% (t =

1.66) for ESKEW.

[Insert Table 11 about here]

We then repeat the portfolio analysis using the residual ST (RES_ST). The results are reported in Table 12. Using RES_ST rather than the raw ST delivers similar results that support our hypothesis. For example, the difference between the risk alpha spread among high RES_ST firms and that among low RES_ST firms is -1.064% (t = -5.23) for TVOL, -0.788% (t = -4.25) for CVOL, -0.742% (t = -3.37) for COV, and -1.348% (t = -4.71) for ESKEW,. While difference of the risk spread between high and low RES_ST is typically smaller than that for raw ST, the difference remains highly significant and economically large using RES_ST.

[Insert Table 12 about here]

In sum, the results in Tables 11 and 12 not only provide out of sample tests for the salience effect but also show that the effect could at least partially explain the existing puzzling risk-return trade-offs in the literature. Our findings suggest that the risk-return relation crucially depends on whether stocks are associated with salient payoffs. The relation between risk and return is negative for stocks with salient upsides and positive for stocks with salient downsides. It worth mentioning that these risk measures were initially motivated by different concepts, and our findings suggest that there is a common underlying mechanism in determining the risk-return relation.

7. Conclusion

Our paper provides a novel psychological explanation for the puzzling risk and return relation in the literature. Our work builds on BGS (2012) who show that investors draw their attention to the stock salient payoffs and exhibit risk-seeking (risk-averse) behavior for stocks with salient upsides (downsides). We hypothesize that high-risk firms are more likely to experience both salient upsides and downsides than low-risk firms. As a result, overpricing (underpricing) of high-risk firms tends to be stronger than that of low-risk firms in the presence of salient upsides (downsides).

We focus on the beta-return relation as our main empirical setting and find that the negative (positive) beta-return relation is much more pronounced among stocks with salient upsides (downsides). To provide support for our hypothesis, we provide several pieces of evidence. First, we find that the salience effect on the beta-return relation is stronger when the market volatility is high since both salient upside and downside returns tend to be more pronounced in such market states. Second, using abnormal search volume as a proxy for retail investor attention, we find that the retail attention is significantly higher among stocks with salient upsides and downsides than non-salient stocks, especially for stocks with salient upsides. We then examine retail investor trading behavior for high and lows beta stocks in the presence salient upsides and downside. Using retail investor buy-sell order imbalanced developed by Boehmer et al. (2021), we find that retail investors demand more high beta stocks than low beta stocks in the presence of salient upsides. By contrast, they demand less high beta stocks in the presence of salient downsides. Finally, we extend our analysis to other risk-return settings and find consistent evidence in all these settings. Overall, our study shows that the salience effect is a key determinant of the risk-return relation.

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Figure 1. Monthly alphas of portfolios double-sorted by beta and salience level

This figure plots the average monthly Fama and French 5-factor (FF-5) alphas (in percentage) on portfolios sequentially sorted into quintiles based on stock beta (BETA) and then into quintiles based on salience theory measure (ST). Detailed variable definitions are described in the appendix. The sample period is from July 1963 to December 2018.

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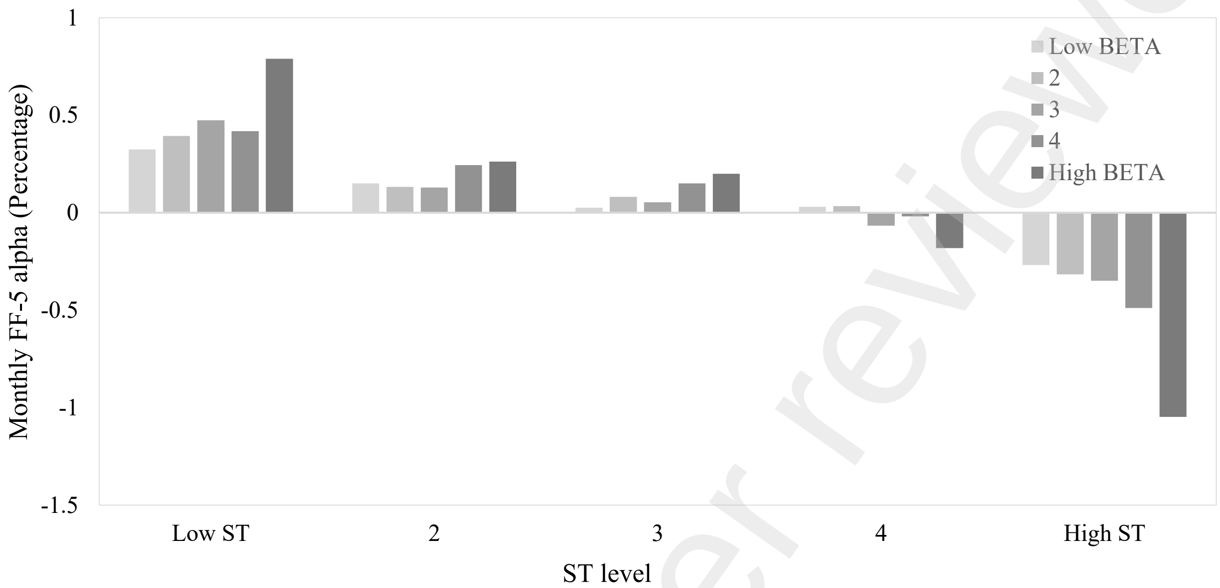

Table 1. Summary characteristics and single sort

This table reports statistics for main variables (Panel A) and correlation between those variables (Panel B) and average percentage returns of beta quintiles (Panel C). The variables include salience theory measure (ST), stock market beta (BETA), log of market value (SIZE), log of book-to-market equity (BM), past 12-month return (RET12), turnover (TURN), institutional ownership (IO), Amihud's (2002) illiquidity (ILLIQ), idiosyncratic volatility (IVOL), Grinblatt and Han's (2005) capital gains overhang, and idiosyncratic skewness (ISKEW). All variables are defined in the appendix. Beta quintiles are formed based on BETA in month *t*. The portfolio is held for 1 month in month *t*+1 and equal-weight. FF-3 and FF-5 alphas are Fama and French (1993) and (2015) risk-adjusted returns respectively. High - Low refers to the high minus low BETA portfolios. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

Panel A: Summary statistics

	Mean	Std	Q1	Median	Q3
ST	0.007	0.031	-0.009	0.004	0.018
BETA	1.089	0.709	0.631	1.000	1.432
SIZE	11.809	2.192	10.175	11.645	13.302
BM	-0.570	0.999	-1.136	-0.483	0.086
RET12	0.168	0.731	-0.186	0.064	0.349
TURN	0.095	0.259	0.017	0.043	0.108
IO	0.387	0.306	0.105	0.337	0.637
ILLIQ	0.302	3.254	0.001	0.008	0.079
IVOL	0.027	0.022	0.013	0.021	0.034
CGO	-0.253	0.776	-0.438	-0.035	0.204
ISKEW	0.245	1.036	-0.273	0.210	0.748

Panel B. Correlation

	ST	BETA	SIZE	BM	RET12	TURN	IO	ILLIQ	IVOL	CGO	ISKEW
ST	1.00										
BETA	0.10	1.00									
SIZE	-0.08	-0.13	1.00								
BM	0.02	-0.12	-0.34	1.00							
RET12	-0.02	0.08	0.08	-0.15	1.00						
TURN	0.13	0.12	0.17	-0.11	0.07	1.00					
IO	-0.10	-0.06	0.69	-0.13	0.01	0.19	1.00				
ILLIQ	0.05	0.01	-0.13	0.07	-0.04	-0.03	-0.08	1.00			
IVOL	0.49	0.26	-0.34	0.00	-0.06	0.17	-0.30	0.15	1.00		
CGO	-0.09	-0.15	0.22	-0.26	0.32	0.01	0.09	-0.07	-0.31	1.00	
ISKEW	0.59	0.05	-0.04	0.01	0.00	0.03	-0.06	0.00	0.16	-0.02	1.00

Panel C. BETA portfolio return

	Excess return	FF-3 alpha	FF-5 alpha
Low BETA	0.649	0.134	0.052
	(3.89)	(2.25)	(0.86)
2	0.818	0.157	0.064
	(3.91)	(2.85)	(1.10)
3	0.856	0.086	0.048
	(3.39)	(1.55)	(0.76)
4	0.808	-0.035	0.061
	(2.73)	(-0.54)	(0.77)
High BETA	0.655	-0.308	0.004
	(1.71)	(-2.42)	(0.04)
High - Low	0.006	-0.442***	-0.048
	(0.02)	(-2.86)	(-0.34)

Table 2. Double sorts on salience and beta

This table reports the monthly average percentage ST value and returns for portfolios sorted on stock beta (BETA) and salience theory measure (ST). In Panels A, B, C and E, the stocks are first sorted into quintiles based on BETA and then into quintiles based on ST. The portfolio is held for 1 month and equal-weight. Panel A reports the average ST within each portfolio. Panels B and C show the average percentage monthly excess returns and alphas with respect to Fama and French (2015) 5-factor (FF-5) respectively. Panel D reports the FF-5 alphas from independent double-sorts on BETA and ST. Panel E reports the value-weighted results of sequential sorts as in Panels B and C. Variables are defined in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	Low BETA	2	3	4	High BETA	High - Low
Panel A. ST						
Low ST	-1.949	-2.104	-2.358	-2.652	-3.035	-1.086***
	(-61.45)	(-69.96)	(-70.13)	(-67.70)	(-64.32)	(-25.96)
2	-0.516	-0.582	-0.622	-0.651	-0.619	-0.103***
	(-32.25)	(-44.18)	(-52.98)	(-47.72)	(-22.21)	(-2.65)
3	0.103	0.230	0.364	0.557	0.910	0.806***
	(6.65)	(18.80)	(30.61)	(31.83)	(25.14)	(17.65)
4	0.829	1.099	1.391	1.781	2.427	1.598***
	(42.72)	(62.73)	(61.24)	(54.21)	(44.55)	(27.96)
High ST	3.177	3.315	3.865	4.574	5.928	2.752***
	(51.83)	(67.03)	(65.67)	(62.21)	(49.61)	(25.80)
High - Low	5.126***	5.419***	6.223***	7.226***	8.963***	3.838***
	(61.26)	(72.35)	(70.48)	(68.07)	(61.49)	(37.72)
Panel B. Excess return						
Low ST	0.990	1.184	1.334	1.206	1.503	0.512*
	(5.37)	(5.20)	(4.99)	(3.81)	(3.79)	(1.91)
2	0.710	0.909	0.923	1.044	0.959	0.249
	(5.12)	(4.62)	(3.99)	(3.72)	(2.65)	(0.90)
3	0.586	0.826	0.862	0.894	0.868	0.282
	(4.32)	(4.33)	(3.71)	(3.27)	(2.38)	(0.99)
4	0.633	0.777	0.755	0.709	0.451	-0.181
	(4.24)	(3.94)	(3.15)	(2.43)	(1.22)	(-0.65)
High ST	0.328	0.395	0.411	0.187	-0.502	-0.830***
	(1.66)	(1.76)	(1.51)	(0.59)	(-1.21)	(-2.92)
High - Low	-0.662***	-0.789***	-0.923***	-1.019***	-2.004***	-1.342***
	(-7.62)	(-8.64)	(-8.77)	(-7.84)	(-10.74)	(-7.31)

	Low BETA	2	3	4	High BETA	High - Low
Panel C. FF-5 alpha						
Low ST	0.325	0.393	0.475	0.418	0.790	0.465***
	(4.06)	(4.82)	(4.56)	(3.15)	(4.49)	(2.65)
2	0.150	0.133	0.129	0.244	0.262	0.112
	(2.44)	(1.95)	(1.80)	(2.64)	(2.08)	(0.78)
3	0.026	0.081	0.053	0.150	0.200	0.174
	(0.40)	(1.28)	(0.80)	(1.75)	(1.82)	(1.27)
4	0.030	0.034	-0.066	-0.018	-0.181	-0.211
	(0.47)	(0.55)	(-0.95)	(-0.21)	(-1.59)	(-1.46)
High ST	-0.269	-0.316	-0.349	-0.489	-1.047	-0.778***
	(-2.70)	(-3.96)	(-3.84)	(-4.36)	(-6.28)	(-4.13)
High - Low	-0.594***	-0.709***	-0.825***	-0.907***	-1.837***	-1.243***
	(-6.22)	(-7.27)	(-6.90)	(-6.07)	(-8.61)	(-6.05)
Panel D. Independent double-sorts FF-5 alpha						
Low ST	0.253	0.362	0.463	0.424	0.756	0.503***
	(2.66)	(4.23)	(4.40)	(3.32)	(4.44)	(2.87)
2	0.144	0.141	0.153	0.243	0.181	0.037
	(2.41)	(2.07)	(2.10)	(2.67)	(1.29)	(0.24)
3	0.041	0.112	0.034	0.164	0.272	0.231*
	(0.73)	(1.84)	(0.50)	(1.79)	(2.27)	(1.69)
4	0.192	-0.060	-0.060	0.026	0.065	-0.126
	(2.53)	(-1.02)	(-0.90)	(0.30)	(0.59)	(-0.85)
High ST	-0.362	-0.333	-0.363	-0.460	-0.843	-0.480***
	(-2.62)	(-3.31)	(-3.58)	(-4.39)	(-6.22)	(-2.85)
High - low	-0.615***	-0.696***	-0.827***	-0.884***	-1.598***	-0.984***
	(-4.12)	(-5.59)	(-6.46)	(-6.34)	(-9.31)	(-5.48)
Panel E. Value-weighted FF-5 alpha						
Low ST	0.122	-0.028	0.118	0.015	0.260	0.138
	(1.26)	(-0.25)	(0.78)	(0.10)	(1.36)	(0.66)
2	-0.021	0.085	0.115	0.159	0.204	0.226
	(-0.27)	(1.02)	(1.06)	(1.32)	(1.15)	(1.10)
3	0.014	0.014	-0.072	0.018	0.139	0.125
	(0.19)	(0.16)	(-0.77)	(0.15)	(0.97)	(0.71)
4	-0.046	-0.150	-0.098	0.077	0.076	0.123
	(-0.55)	(-1.97)	(-1.03)	(0.58)	(0.38)	(0.59)
High ST	-0.086	-0.094	-0.248	0.160	-0.605	-0.519**
	(-0.69)	(-0.77)	(-1.96)	(1.03)	(-3.26)	(-2.16)
High - Low	-0.208	-0.066	-0.366*	0.144	-0.865***	-0.657**
	(-1.29)	(-0.42)	(-1.73)	(0.72)	(-3.39)	(-2.26)

Table 3. Fama-MacBeth regressions

This table reports results of Fama-MacBeth (1973) regression of monthly returns on lagged variables. The time-series average of the estimated coefficient, *t*-statistics, the number of observations and adjusted R^2 are reported. BETA is the stock market beta. ST is the salience theory measure. We construct salience theory low (ST_LOW) and high (ST_HIGH) dummy variables for each stock if its ST belongs bottom and top quintile respectively. SIZE is the log of market value. BM is the log of book-to-market equity. RET1 is the last month return, RET12 is the cumulative return over the past year with a one-month gap. RET36 is the cumulative return over the past three years with a one-year gap. ILLIQ is the monthly Amihud's (2002) illiquidity measure. TURN is the turnover as monthly trading volume divided by the number of shares outstanding. Detailed variable constructions are shown in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. Independent variables are winsorized at 5% and 95%. The regression coefficients are reported in percentage. The intercept of the regression is not reported. The sample period is from July 1963 to December 2018. All t-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%).

	(1)	(2)	(3)	(4)	(5)	(6)
BETA	0.123	-0.003	0.186	0.189	0.039	0.060
	(0.72)	(-0.02)	(1.07)	(1.09)	(0.30)	(0.49)
ST	-19.764***	-5.537***		-5.441**		8.447***
	(-13.34)	(-4.55)		(-2.31)		(3.25)
ST_HIGH			0.020		0.236*	
			(0.14)		(1.92)	
ST_LOW			0.141		-0.271***	
			(1.47)		(-3.27)	
BETA×ST_HIGH			-0.582***		-0.399***	
			(-6.32)		(-4.31)	
BETA×ST_LOW			0.257***		0.227***	
			(3.25)		(3.05)	
BETA×ST				-11.935***		-11.300***
				(-5.74)		(-5.35)
SIZE		-0.065*			-0.067**	-0.060*
		(-1.96)			(-2.06)	(-1.82)
BM		0.313***			0.310***	0.313***
		(5.44)			(5.42)	(5.46)
RET1		-5.218***			-5.469***	-5.302***
		(-11.09)			(-12.03)	(-11.18)
RET12		1.072***			1.071***	1.075***
		(6.34)			(6.34)	(6.36)
RET36		-0.134*			-0.133*	-0.133*
		(-1.96)			(-1.95)	(-1.95)
ILLIQ		0.858			0.863	0.856
		(1.60)			(1.61)	(1.60)
TURN		2.235*			2.268*	2.101
		(1.73)			(1.79)	(1.64)
No. of Obs.	1,876,621	1,703,521	1,876,621	1,876,621	1,703,521	1,703,521
Adj. R^2	2.589%	6.300%	2.773%	2.677%	6.408%	6.385%

Table 4. Market states, salience and beta

This table reports the monthly average percentage Fama and French 5-factor alphas for portfolios sorted on BETA and salience theory measure (ST) during periods of high and high and low market volatility. High and low market volatility is as classified based on the median of the volatility of 6-month value-weighted market returns during the whole sample. We sort stocks into quintiles based on BETA and then into quintiles based on ST. The portfolio is held for 1 month and equally weighted. Detailed variable definitions are described in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	High volatility			Low volatility			High volatility – Low volatility		
	Low BETA	High BETA	High - Low	Low BETA	High BETA	High - Low	Low BETA	High BETA	High - Low
Low ST	0.347	1.363	1.016***	0.316	0.270	-0.046	0.031	1.093	1.062***
	(2.56)	(4.97)	(3.73)	(3.13)	(1.37)	(-0.23)	(0.18)	(3.38)	(3.23)
2	0.111	0.615	0.503**	0.193	-0.086	-0.279*	-0.082	0.701	0.782***
	(1.18)	(2.93)	(2.17)	(2.42)	(-0.68)	(-1.76)	(-0.66)	(2.95)	(2.86)
3	-0.080	0.399	0.479**	0.126	0.023	-0.103	-0.206	0.376	0.582**
	(-0.78)	(2.22)	(2.21)	(1.66)	(0.19)	(-0.65)	(-1.66)	(1.75)	(2.23)
4	-0.072	-0.229	-0.157	0.134	-0.152	-0.286*	-0.206	-0.077	0.129
	(-0.76)	(-1.21)	(-0.67)	(1.72)	(-1.19)	(-1.75)	(-1.74)	(-0.35)	(0.46)
High ST	-0.332	-1.145	-0.813***	-0.201	-0.972	-0.771***	-0.132	-0.174	-0.042
	(-2.07)	(-4.43)	(-2.80)	(-1.81)	(-4.80)	(-3.39)	(-0.71)	(-0.55)	(-0.12)
High - Low	-0.679***	-2.509***	-1.830***	-0.516***	-1.242***	-0.725***	-0.162	-1.267***	-1.104***
	(-4.77)	(-8.04)	(-5.84)	(-4.54)	(-5.00)	(-2.99)	(-0.95)	(-3.43)	(-2.92)

Table 5. Evidence from retail investor attention and trading activity

This table reports the evidence from retail investor attention and retail buy and sell order imbalance. The retail attention is proxied by abnormal Google search volume (ASVI) following Da, Engelberg and Gao(2011) and the retail buy-sell order imbalance (RBSI) is based on the sub-penny price improvement by Boehmer et al. (2021). Panel A reports the ASVI doubled sorted by beta and salience portfolios. Panel B reports results of Fama-MacBeth (1973) regression of monthly RBSI on contemporaneous variables. The time-series average of the estimated coefficient, *t*-statistics, the number of observations and adjusted R^2 are reported. BETA is the stock market beta. ST is the salience theory measure. ST_C is the categorical variable of ST ranging from 0 to 4, with 0 (4) being stocks with the lowest (highest) ST. The control variables are the same as those in Table 3. Detailed variable constructions are shown in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. Independent variables are winsorized at 5% and 95%. The regression coefficients are reported in percentage. The intercept of the regression is not reported. The sample period is from July 1963 to December 2018. All t-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%).

	Low BETA	2	3	4	High BETA	High - Low
Panel A. Abnormal Google search volume (ASVI)						
Low ST	1.326	1.566	1.199	1.682	1.450	0.125
	(3.98)	(4.39)	(3.21)	(5.07)	(3.03)	(0.28)
2	-0.042	-0.610	-1.018	-0.899	-1.462	-1.420***
	(-0.13)	(-2.08)	(-3.03)	(-2.49)	(-3.41)	(-3.34)
3	-0.307	-0.442	-0.326	-0.955	-2.003	-1.696***
	(-1.01)	(-1.47)	(-0.92)	(-2.38)	(-4.63)	(-4.04)
4	0.324	-0.306	0.412	-0.177	-0.212	-0.536
	(0.87)	(-1.02)	(1.06)	(-0.47)	(-0.49)	(-1.22)
High ST	3.328	3.352	4.231	4.667	6.750	3.421***
	(8.75)	(9.23)	(11.69)	(9.31)	(10.46)	(6.06)
High - Low	2.003***	1.786***	3.031***	2.984***	5.299***	3.297***
	(5.06)	(4.87)	(9.70)	(7.01)	(10.11)	(5.25)

Panel B. Retail investor buy and sell order imbalance (RBSI)		
	(1)	(2)
BETA	-0.014***	-0.027***
	(-2.84)	(-3.99)
ST	1.358***	
	(2.76)	
BETA×ST	0.683**	
	(2.57)	
ST_C		0.002
		(0.43)
BETA×ST_C		0.008***
		(2.91)
Control	Yes	Yes
No. of Obs.	262,737	262,737
Adj. R^2	3.398%	3.797%

Table 6. Robustness checks

This table reports a series of robustness Fama-MacBeth regression tests. We take the regression specification in Column (4) in Table 3. For brevity, only the coefficients of the interaction term of beta (BETA) and salience theory measure (ST), the number of observations and adjusted R^2 are reported. In models (1), (2) and (3), we use shrinkage adjustment BETA, non-synchronous adjustment BETA and Frazzini and Pedersen's (2014) BETA respectively in the analysis. In models (4) and (5), we only include NYSE/AMEX and NASDAQ stocks respectively. In model (6), we do not winsorize independent variables in the analysis. In Model (7), we only include stocks with a price at least \$5 instead of \$1. Model (8) performs the weighted least squared regression. Model (9) excludes top illiquid 10% stocks based on Amihud (2002) illiquidity measure. Model (10) uses the residual BETA instead of BETA. The residual BETA is computed as the residual of cross-sectional regression of BETA on idiosyncratic skewness. Models (11) and (12) perform the regressions in two equal subperiods from July 1963 to December 1989 and from January 1990 to December 2018 respectively. Models (13), (14) and (15) measure ST over 3- , 6- and 12-month estimation windows rather than 1-month. Models (16) and (17) measure ST using the weekly frequency data over the last month and using the monthly frequency over 12-month window respectively. Model (18) estimates ST by deleting the last day's stock return over the analyzed month to alleviate the impact of bid-ask bounce. All variables are defined in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least \$1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. Independent variables are winsorized at 5% and 95%. The regression coefficients are reported in percentage. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Shrinkage adjustment BETA	Non-synchronous adjustment BETA	FP (2014) BETA	NYSE/AMEX	NASDAQ	No winsorization	Price >= 5	WLS	Exclude illiquid firms
BETA×ST	-19.043***	-16.237***	-4.056**	-9.171***	-6.230***	-7.745***	-6.218***	-11.848***	-8.907***
	(-5.23)	(-5.04)	(-2.14)	(-3.37)	(-3.47)	(-3.73)	(-2.69)	(-5.35)	(-4.10)
Adj. R^2	6.420%	6.330%	6.689%	7.471%	4.021%	6.196%	7.012%	6.385%	6.675%
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
	Residual BETA	July 1963 – Dec. 1989	Jan. 1990 – Dec. 2018	ST over 3-month window	ST over 6-month window	ST over 12-month window	ST using weekly frequency	ST using monthly frequency	ST deleting the last day return
BETA×ST	-11.125***	-15.958***	-4.303**	-4.163***	-2.861***	-2.183***	-11.073***	-1.887***	-10.802***
	(-5.22)	(-4.86)	(-2.44)	(-3.56)	(-3.05)	(-2.76)	(-4.14)	(-2.73)	(-4.94)
Adj. R^2	6.384%	8.006%	4.905%	6.416%	6.449%	6.495%	6.355%	6.387%	6.374%

Table 7. Fama-MacBeth regression controlling for lottery preference

This table reports the results of Fama-MacBeth regressions after controlling for the effect of lottery preference. We take the regression specification in column (4) of Table 3 and add three lottery preference proxies and their interactions with stock beta (BETA) to the regressions. The three lottery proxies are maximum daily return in a month (MAX), idiosyncratic skewness (IKEW) and expected skewness (ESKEW). The time-series average of the estimated coefficient, *t*-statistics, the number of observations and adjusted R^2 are reported. Detailed variable definitions are shown in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. Independent variables are winsorized at 5% and 95%. The regression coefficients are reported in percentage. The intercept of the regression is not reported. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%).

	(1)	(2)	(3)	(4)	(5)	(6)
BETA	0.392**	0.289*	0.010	0.045	0.028	0.047
	(2.56)	(1.84)	(0.08)	(0.37)	(0.13)	(0.22)
ST		14.947***		4.646		1.909
		(5.27)		(1.37)		(0.81)
BETA×ST		-7.568***		-13.355***		-4.476***
		(-3.07)		(-4.80)		(-2.83)
MAX	-1.633	-5.624***				
	(-0.99)	(-2.98)				
BETA×MAX	-5.506***	-3.555**				
	(-4.37)	(-2.35)				
ISKEW			0.204***	0.021		
			(5.11)	(0.44)		
BETA×ISKEW			-0.163***	0.143***		
			(-4.37)	(3.16)		
ESKEW					-0.561***	-0.560***
					(-2.62)	(-2.64)
BETA×ESKEW					0.059	0.084
					(0.41)	(0.59)
Control	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	1,703,521	1,703,521	1,702,222	1,702,222	1,227,949	1,227,949
Adj. R^2	6.567%	6.670%	6.284%	6.424%	4.918%	5.024%

Table 8. Fama-MacBeth regression controlling for salience induced mispricing and limits to arbitrage

This table reports the results of Fama-MacBeth regressions after controlling for salience induced mispricing and limits to arbitrage. We take the regression specification in column (4) of Table 3 and add three limits to arbitrage proxies and their interactions with salience (ST) to the regressions. The three limits to arbitrage proxies are institutional ownership (IO), Amihud's (2002) illiquidity measure (ILLIQ) and idiosyncratic volatility (IVOL). The time-series average of the estimated coefficient, *t*-statistics, the number of observations and adjusted R^2 are reported. Detailed variable definitions are shown in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. Independent variables are winsorized at 5% and 95%. The regression coefficients are reported in percentage. The intercept of the regression is not reported. All t-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%).

	(1)	(2)	(3)	(4)	(5)	(6)
BETA	-0.016	0.019	-0.011	0.045	0.058	0.109
	(-0.13)	(0.14)	(-0.09)	(0.36)	(0.51)	(0.94)
ST	-10.092***	-1.900	1.401	12.998***	12.709***	20.554***
	(-4.66)	(-0.65)	(1.04)	(4.85)	(5.16)	(6.50)
BETA×ST		-6.298***		-9.610***		-9.163***
		(-3.67)		(-4.70)		(-4.44)
IOR	1.374***	1.371***				
	(6.58)	(6.59)				
IOR×ST	22.154***	20.163***				
	(5.85)	(5.35)				
ILLIQ			1.199**	1.194**		
			(2.24)	(2.24)		
ILLIQ×ST			-29.869*	-30.001*		
			(-1.83)	(-1.80)		
IVOL					-25.305***	-26.175***
					(-8.11)	(-8.40)
IVOL×ST					-334.27***	-225.42**
					(-3.56)	(-2.36)
Control	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	1,389,068	1,389,068	1,703,521	1,703,521	1,703,521	1,703,521
Adj. R^2	5.138%	5.195%	6.404%	6.481%	6.576%	6.647%

Table 9. Fama-MacBeth regression controlling for other mechanisms

This table reports the results of Fama-MacBeth regressions by taking account of reference-dependent preference, coskewness risk and disagreement effect. We take the regression specification in column (4) of Table 3 and additional variables into the regression. Columns (1) and (2) take account of reference-dependent preference proxied by the capital gains overhang (CGO) as in Grinblatt and Han (2005). Column (3) takes account of coskewness risk (COSKEW) as in Harvey and Siddique (2000). Columns (4) and (5) perform the regression analysis in non-speculative and speculative stocks samples. Detailed variable definitions are shown in the appendix. The speculativeness of a particular stock is defined as the stock beta by idiosyncratic variance following Hong and Sraer (2016). Then stocks are classified into speculative and non-speculative samples each month if the speculativeness is above and below the median of value respectively. The time-series average of the estimated coefficient, t-statistics, the number of observations and adjusted R^2 are reported. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. Independent variables are winsorized at 5% and 95%. The regression coefficients are reported in percentage. The intercept of the regression is not reported. All t-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%).

	(1)	(2)	(3)	(4)	(5)
				Non-speculative Stocks	Speculative Stocks
BETA	-0.106	-0.057	0.062	-0.069	-0.019
	(-0.79)	(-0.43)	(0.50)	(-0.55)	(-0.13)
ST	-4.112***	9.714***	8.471***	7.365***	12.190***
	(-3.35)	(3.46)	(3.26)	(2.62)	(3.42)
BETA×ST		-11.376***	-11.291***	-10.478***	-7.926***
		(-4.77)	(-5.35)	(-4.40)	(-2.79)
CGO	-0.194*	-0.162			
	(-1.79)	(-1.52)			
BETA×CGO	0.200**	0.175*			
	(2.11)	(1.87)			
COSKEW			0.071		
			(1.25)		
Control	Yes	Yes	Yes	Yes	Yes
No. of obs.	1,262,923	1,262,923	1,702,222	846,803	856,121
Adj. R^2	7.148%	7.245%	6.435%	5.638%	7.720%

Table 10. Double sorts on residual salience and beta

This table reports the monthly average percentage returns for portfolios sorted on stock beta (BETA) and residual salience theory measure (RES_ST). RES_ST is estimated from Eq. (8) as the residual from a cross-sectional regression of salience (ST) on capital gains overhang (CGO), maximum daily return (MAX), idiosyncratic skewness (ISKEW), coskewness (COSKEW) and past one-month return (RET1). The stocks are first sorted into quintiles based on BETA and then into quintiles based on RES_ST. The portfolio is held for 1 month and equal-weight. Panels A and B show the average percentage monthly excess returns and alphas with respect to Fama and French (2015) 5-factor (FF-5). All variables are defined in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	Low BETA	2	3	4	High BETA	High - Low
Panel A. Excess return						
Low RES_ST	0.692	0.852	1.048	1.056	1.209	0.517*
	(4.51)	(4.16)	(4.27)	(3.59)	(3.20)	(1.85)
2	0.666	0.860	0.945	0.944	0.923	0.257
	(4.77)	(4.47)	(4.19)	(3.46)	(2.66)	(0.96)
3	0.694	0.848	0.852	0.924	0.852	0.159
	(4.94)	(4.49)	(3.81)	(3.42)	(2.41)	(0.57)
4	0.694	0.862	0.802	0.896	0.763	0.070
	(4.87)	(4.33)	(3.45)	(3.13)	(2.01)	(0.24)
High RES_ST	0.549	0.688	0.760	0.474	0.019	-0.530**
	(3.16)	(3.10)	(2.79)	(1.42)	(0.04)	(-1.99)
High - Low	-0.143*	-0.164**	-0.289***	-0.582***	-1.190***	-1.047***
	(-1.84)	(-2.11)	(-2.93)	(-4.85)	(-6.31)	(-5.46)
Panel B. FF-5 alpha						
Low RES_ST	0.070	0.091	0.207	0.265	0.404	0.334*
	(1.08)	(1.32)	(2.47)	(2.39)	(2.42)	(1.88)
2	0.093	0.096	0.116	0.117	0.176	0.083
	(1.37)	(1.46)	(1.54)	(1.33)	(1.49)	(0.58)
3	0.117	0.060	0.051	0.107	0.176	0.059
	(1.76)	(0.88)	(0.78)	(1.21)	(1.46)	(0.40)
4	0.119	0.089	-0.045	0.142	0.031	-0.089
	(1.75)	(1.40)	(-0.66)	(1.43)	(0.25)	(-0.63)
High RES_ST	-0.047	-0.075	-0.055	-0.265	-0.660	-0.613***
	(-0.56)	(-0.93)	(-0.58)	(-2.31)	(-3.30)	(-2.87)
High - Low	-0.117	-0.166**	-0.262***	-0.530***	-1.065***	-0.948***
	(-1.49)	(-2.18)	(-2.63)	(-4.31)	(-4.77)	(-4.08)

Table 11. Double sorts on residual salience and risk: other settings

This table reports the monthly average percentage Fama and French 5-factor (FF-5) alphas for portfolios sorted on one of the risk proxies and salience (ST). The stocks are first sorted into quintiles based on one of the risk proxies and then into quintiles based on ST. TVOL is the standard deviation of returns over the month. CVOL is the cash flow volatility computed as the standard deviation of cash flow from operations over the past 5 years. COV is the analyst coverage as the number of analysts following the firm in the last year. Detailed variable definitions are shown in the appendix. ESKEW is expected skewness. The portfolio is held for 1 month and equal weight. High risk firms are those with high TVOL, and high CVOL, low COV and high ESKEW. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report results for low, third and high ST quintiles and asterisks for the hedged portfolios.

	Low Risk	2	3	4	High Risk	High - Low
Panel A. TVOL						
Low ST	0.057	0.445	0.459	0.605	0.533	0.476**
	(0.77)	(5.17)	(4.45)	(4.67)	(2.92)	(2.51)
3	-0.067	0.138	0.232	0.132	-0.358	-0.291*
	(-1.08)	(2.21)	(3.35)	(1.29)	(-2.42)	(-1.78)
High ST	0.010	-0.084	0.045	-0.234	-1.188	-1.199***
	(0.17)	(-1.34)	(0.73)	(-2.88)	(-6.86)	(-6.22)
High - Low	-0.047	-0.529***	-0.415***	-0.839***	-1.721***	-1.675***
	(-0.66)	(-5.70)	(-3.89)	(-5.86)	(-7.95)	(-7.72)
Panel B. CVOL						
Low ST	0.355	0.438	0.519	0.581	0.592	0.237*
	(4.42)	(4.61)	(4.67)	(4.63)	(4.15)	(1.85)
3	0.054	0.108	0.152	0.136	0.194	0.141
	(0.92)	(1.76)	(2.39)	(1.81)	(1.80)	(1.16)
High ST	-0.254	-0.385	-0.403	-0.397	-0.773	-0.519***
	(-3.67)	(-4.59)	(-4.42)	(-3.29)	(-4.59)	(-3.25)
High - Low	-0.610***	-0.823***	-0.922***	-0.977***	-1.366***	-0.756***
	(-6.02)	(-7.30)	(-7.29)	(-6.08)	(-7.23)	(-4.55)
Panel C. COV						
Low ST	-0.077	0.135	-0.075	0.042	0.181	0.259
	(-0.44)	(0.78)	(-0.51)	(0.26)	(1.14)	(1.45)
3	-0.049	-0.055	-0.036	0.072	0.354	0.403***
	(-0.59)	(-0.62)	(-0.44)	(0.68)	(3.34)	(3.34)
High ST	0.043	-0.221	-0.331	-0.487	-0.757	-0.800***
	(0.32)	(-1.78)	(-2.51)	(-3.31)	(-4.85)	(-4.25)
High - Low	0.120	-0.355**	-0.256	-0.529***	-0.939***	-1.059***
	(0.72)	(-2.22)	(-1.44)	(-2.75)	(-4.91)	(-5.18)
Panel D. ESKEW						
Low ST	0.041	0.052	0.189	0.389	0.516	0.475*
	(0.35)	(0.43)	(1.33)	(2.09)	(1.72)	(1.66)
3	0.120	-0.007	0.069	0.271	0.530	0.410*
	(1.70)	(-0.09)	(0.75)	(1.97)	(2.37)	(1.76)
High ST	-0.065	-0.054	-0.267	-0.444	-0.946	-0.881***
	(-0.58)	(-0.58)	(-2.60)	(-2.90)	(-3.39)	(-2.91)
High - Low	-0.106	-0.106	-0.456***	-0.833***	-1.463***	-1.356***
	(-0.71)	(-0.68)	(-2.65)	(-4.24)	(-5.13)	(-5.14)

Table 12. Double sorts on residual salience and risk: other settings

This table reports the monthly average percentage Fama and French 5-factor (FF-5) alphas for portfolios sorted on risk proxy and residual salience (RES_ST). RES_ST *is* estimated from Eq. (8) as the residual from a cross-sectional regression of salience (ST) on capital gains overhang (CGO), maximum daily return (MAX), idiosyncratic skewness (ISKEW) and coskewness (COSKEW) and past one-month return (RET1). The stocks are first sorted into quintiles based on one of the risk proxies and then into quintiles based on RES_ST. TVOL is the standard deviation of returns over the month. CVOL is the cash flow volatility computed as the standard deviation of cash flow from operations over the past 5 years. COV is the analyst coverage as the number of analysts following the firm in the last year. ESKEW is expected skewness. The portfolio is held for 1 month and equal weight. High risk firms are those with high TVOL, and high CVOL, low COV and high ESKEW. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	Low Risk	2	3	4	High Risk	High - Low
Panel A. TVOL						
Low ST	-0.033	0.169	0.204	0.376	0.278	0.311
	(-0.58)	(2.70)	(2.58)	(3.45)	(1.58)	(1.62)
3	0.064	0.134	0.175	0.083	-0.304	-0.368**
	(1.03)	(2.01)	(2.49)	(0.92)	(-2.01)	(-2.20)
High ST	-0.019	0.171	0.324	-0.042	-0.772	-0.753***
	(-0.33)	(2.55)	(3.84)	(-0.47)	(-4.03)	(-3.69)
High - Low	0.015	0.002	0.120	-0.418***	-1.050***	-1.064***
	(0.28)	(0.03)	(1.60)	(-3.94)	(-5.34)	(-5.23)
Panel B. CVOL						
Low ST	0.132	0.166	0.219	0.132	0.305	0.173
	(2.06)	(2.11)	(2.40)	(1.34)	(2.34)	(1.31)
3	0.009	0.103	0.075	0.075	0.122	0.113
	(0.15)	(1.77)	(1.07)	(0.92)	(1.21)	(0.96)
High ST	-0.015	-0.053	-0.175	-0.200	-0.630	-0.615***
	(-0.20)	(-0.59)	(-1.76)	(-1.51)	(-3.45)	(-3.34)
High - Low	-0.147*	-0.219***	-0.394***	-0.331**	-0.935***	-0.788***
	(-1.87)	(-2.58)	(-4.01)	(-2.54)	(-5.27)	(-4.25)
Panel C. COV						
Low ST	-0.140	-0.055	-0.117	-0.065	0.103	0.243
	(-1.09)	(-0.52)	(-0.99)	(-0.50)	(0.64)	(1.43)
3	-0.018	-0.031	-0.081	0.063	0.435	0.453***
	(-0.19)	(-0.32)	(-0.86)	(0.59)	(3.90)	(3.32)
High ST	0.014	-0.213	-0.172	-0.254	-0.484	-0.499**
	(0.11)	(-1.57)	(-1.32)	(-1.43)	(-2.71)	(-2.56)
High - Low	0.154	-0.158	-0.056	-0.189	-0.588***	-0.742***
	(1.33)	(-1.30)	(-0.43)	(-1.16)	(-2.95)	(-3.37)
Panel D. ESKEW						
Low ST	-0.016	0.011	0.099	0.439	0.495	0.511*
	(-0.19)	(0.11)	(0.81)	(2.81)	(1.97)	(1.94)
3	0.010	0.024	0.097	0.377	0.252	0.242
	(0.11)	(0.28)	(1.10)	(2.67)	(1.26)	(1.10)
High ST	0.077	0.093	-0.155	-0.191	-0.760	-0.837***
	(0.82)	(0.92)	(-1.36)	(-1.15)	(-2.63)	(-2.83)
High - Low	0.093	0.082	-0.254**	-0.619***	-1.255***	-1.348***
	(0.86)	(0.71)	(-2.07)	(-4.13)	(-4.70)	(-4.71)

Appendix. Variable description and construction

Variable Name	Description and Construction
ST	The salience theory value is calculated as the covariance between salience weights and daily returns as follows: $$ST_{i,t} = cov[\omega_{i,s,t}, r_{i,s,t}]$$ where $\omega_{i,s} = \frac{\delta^{k_{i,s}}}{\sum_{s'} \pi_{s'} \delta^{k_{i,s}}}$ and the value δ equals to 0.7. $k_{i,s}$ is the salience rank ranging from 1 for the most salient to *S* for the least salient. The value *S* is the number of trading days in the analyzed month. The salience of the daily payoff is based on $\sigma(r_{i,s}, r_{m,s}) = \frac{\lvert r_{i,s} - r_{m,s} \rvert}{\lvert r_{i,s} \rvert + \lvert r_{m,s} \rvert + \theta}$ where θ equals to 0.1, $r_{i,s}$ is the stock *i*'s return on day *s* and $r_{m,s}$ is the market return of equal-weighted CRSP index on day *s*.
BETA	Stock market beta is the slope coefficient of regression of monthly stock excess returns on market excess return over a 60-month rolling window with a minimum of 24 months.
SIZE	The firm size is measured as the log of the firm's market capitalization, which is calculated as the number of shares outstanding multiplied by the share price at the end of month.
BM	The book to market ratio is the log of firm's book value of equity divided by its market capitalization, where we follow Fama and French (2008).
RET1	The firm's stock return over the past month.
RET12	The firm's cumulative stock return over the past 12 months skipping the most recent month.
RET36	The firm's cumulative stock return over the past 13- to 36-month.
ILLIQ	Amihud's (2002) illiquidity is computed as the average ratio of the daily stock's absolute return to its dollar volume in the month.
TURN	Stock turnover is defined as the number of shares traded each month divided by the number of shares outstanding at the end of that month. To address the double counting issue of NASDAQ stocks, we follow Gao and Ritter (2010) and scale down Nasdaq volume by 2 prior to February 1, 2001, by 1.8 from February 1, 2001 to the end of December of 2002, by 1.6 during 2002, and by 1 thereafter to make it approximately comparable to NYSE volume.
IVOL	Idiosyncratic volatility is the standard deviation of residuals from regression on daily return on Fama and French (1993) three factors in each month with a minimum of 15 observations.
IO	The institutional ownership is computed as the number of shares held by all institutional investors in the quarter scaled by the number of shares outstanding.
MAX	The maximum daily return in each month
ISKEW	Idiosyncratic skewness is estimated based the regression residual using the Fama and French (1993) three factors on day d for stock *i* as follows $$iv_{i,t} = (\frac{1}{N(t)} \sum_{d \in S(t)} \varepsilon_{i,d}^2)^{1/2}$$ $$is_{i,t} = \frac{1}{N(t)} \frac{\sum_{d \in S(t)} \varepsilon_{i,d}^3}{iv_{i,t}^3}$$

	where $is_{i,t}$ and $iv_{i,t}$ are the historical estimates of skewness and idiosyncratic volatility for firm *i* relative to Fama and French (1992) three-factor model using daily data for all days in *S(t)* (past 60 months).
ESKEW	Expected idiosyncratic skewness is computed following Boyer et al. (2010) (Model 6 of Table 2). We first run cross-sectional regression at the end of each month *t* to estimate the coefficient of each variable: $is_{i,t} = \beta_{0,t} + \beta_{1,t} is_{i,t-T} + \beta_{2,t} iv_{i,t-T} + \lambda'_{t\,t} X_{i,t-T} + \varepsilon_{i,t}$ where $is_{i,t}$ and $iv_{i,t}$ are the historical estimates of skewness and idiosyncratic volatility for firm *i* relative to Fama and French (1992) three-factor model using daily data over past 60 months. $X_{i,t-T}$ is a vector of firm-specific variables including momentum as the cumulative returns over months *t*-72 to *t*-60, turnover as the average daily turnover in month *t*-60, the small and medium-size dummies, where firms are categorized into terciles based on market capitalization, the industry dummies for 16 of the 17 industries based on Fama and French industry definition, and the NASDAQ dummy. Next, the expected idiosyncratic skewness for each firm at the end of each month *t* is calculated by using the coefficients of the variables from the first step: $ESKEW_{i,t} = E_t[is_{i,t+60}] = \beta_{0,t} + \beta_{1,t} is_{i,t} + \beta_{2,t} iv_{i,t} + \lambda'_{t\,t} X_{i,t}$.
ASVI	The abnormal Google search volume is constructed following Da et al. (2011). The data can be accessed at trends.google.com. ASVI is computed as the log of Google search volume for a stock during the month minus the log of median SVI during the previous six months.
RBSI	Retail buy and sell order imbalance is constructed following Boehmer et al. (2021). For each stock, the daily RBSI calculated as the number shares as classified as retail investor buy minus the number of shares as classified as retail investor sell and then scaled by the number of shares outstanding. The monthly RBSI is the average of daily RBSI within that month.
CGO	Capital gains overhang. The capital gains overhang variable in month *t*, following Grinblatt and Han (2005): $$\mathrm{CGO_t} = \frac{\mathrm{P_{t-2} - RP_{t-1}}}{\mathrm{P_{t-2}}}$$ where P_{t-2} is the stock price at the end of the second-to-last week of month *t*, and $$\mathrm{RP_t} = \frac{1}{\mathrm{K}} \sum_{\mathrm{n}=1}^{260} \left(\mathrm{V_{t-n}} \prod_{\tau}^{\mathrm{n}-1} (1 - \mathrm{V_{t-n+\tau}}) \right) \mathrm{P_{t-1-n}}$$ where P_t is the stock price at the end of week *t*; V_t is stock turnover in week *t*; K is a constant that makes the weights on past prices sum to one. The weight is the term in brackets, which is the probability that the stock was last bought at week *t*-n and has not been traded since. Weekly turnover is calculated as weekly trading volume divided by the number of shares outstanding. To be included in the sample, a stock must have at least 100 weeks of non-missing data in the previous 5 years. At last, the monthly CGO is the average of weekly CGO within each month.
COSKEW	Coskewness risk is the slope coefficient from regression of CAPM residual returns on squared market excess returns.
RES_BETA	Residual BETA is estimated in each month from a cross-sectional regression of

	BETA on idiosyncratic skewness. RES_BETA is the residual from the regression.
RES_ST	Residual ST is estimated in each month from a cross-sectional regression of ST on a set of variables below: $ST_{i,t} = \gamma_0 + \gamma_1 CGO_{i,t} + \gamma_2 MAX_{i,t} + \gamma_3 ISKEW_{i,t} + \gamma_4 COSKEW_{i,t} + \gamma_5 RET1_{i,t} + \varepsilon_{i,t}$ where for stock *i*, ST is the salience theory measure in month *t*, CGO is the capital gains overhang in month *t*, MAX is maximum daily return in month *t*, ISKEW is idiosyncratic volatility in month *t*. COSKEW is coskewness beta in month *t*. RET1 is the stock return in month *t*. RES_ST is the residual from the regression.

Internet Appendix: BGS (2012) and asset pricing

In Bordalo, Gennaioli, and Shleifer (2012). Salience Theory of Choice Under Risk. NBER Working Paper 16387, model of salience is included in a standard asset pricing model. We sketch the model here.

Suppose that the investor values consumption according to a concave utility function $u(c)$ ($u' > 0$; $u'' \leq 0$)) and there is no time discounting. Each unit of asset $i \in [0; 1]$ costs p_i units of current consumption and yields at $t = 1$ a dividend x_s^i in state $s \in S$. The salience of a state s is denoted by $\sigma(x_s^i, x_s)$. The same as the standard utility maximization, except for the fact that a deviation along the holding of each asset j is evaluated using the asset-specific decision weights $\omega_{s,j}$. the first order condition for the investor is:

$p_i \cdot u'(c_0) = E[\omega_s^i \cdot x_s^i \cdot u'(c_{1,s})]$ for all $i \in [0,1]$, where c_0 and $c_{1,s}$ are the consumption levels at each time and in each state. It departs from the standard Euler equation because the payoff x_s^i of the asset is weighted by its salience ω_s^i. In general equilibrium, $p_i = \frac{E[x_s^i]}{R} + \frac{cov[\omega_s^i, x_s^i]}{R}$, for all $i \in [0,1]$, where $R = {u'(c_0)}/{u'_{(c_1)}}$ is the return on the riskless asset. The first term on the right hand side of the equation is the price prevailing when the agent is fully rational (i.e. when $\delta = 1$): in the absence of aggregate risk, the investor is risk neutral at the margin, valuing each asset at its expected discounted dividend. Indeed, for a diversified investor, standard risk aversion is second order, leaving prices unaffected. Relative to an Expected Utility maximizer, the local thinker over or under values an asset by the second term on the right hand side, which increases in the covariance between the asset's payoffs and their salience. Specifically, the local thinker overvalues an asset - exhibiting risk-seeking behavior - when the asset's highest payoffs are salient, while he undervalues an asset, exhibiting risk-averse behavior, when the asset's lowest payoffs are salient. By shaping the agent's focus on specific asset payoffs, salience creates a first order source of risk attitudes. In terms of return, $E(r_i) - R = -cov(\omega_s^i, r_{i,s})$. In case of $\delta < 1$, the investor overweights salient payoffs at the expenses of the non salient ones. When the lowest payoffs of an asset are the salient ones, the covariance is negative, the investor focuses on downside risk and demand a positive risk premium. When the highest payoffs are the salient ones, the covariance is positive, the investors focus on the asset's upside's potential and the risk premium is negative.

Internet Appendix: Additional Tables

Table A1. Sequential double-sorts on salience and beta

This table reports the monthly average percentage alphas with respect to Fama and French (2015) 5-factor (FF-5) for portfolios sorted on salience theory measure (ST) and stock beta (BETA). The stocks are first sorted into quintiles based on ST and then into quintiles based on BETA. The portfolio is held for 1 month and equal-weight. The variables are defined in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	Low BETA	2	3	4	High BETA	High - Low
Low ST	0.199	0.374	0.386	0.404	0.702	0.503***
	(2.34)	(3.77)	(3.43)	(2.85)	(3.90)	(2.85)
2	0.108	0.089	0.099	0.100	0.146	0.037
	(1.57)	(1.39)	(1.43)	(1.25)	(1.25)	(0.28)
3	0.058	0.080	0.086	0.109	0.197	0.138
	(0.90)	(1.29)	(1.43)	(1.58)	(1.91)	(1.09)
4	0.199	-0.013	0.046	0.064	0.072	-0.127
	(2.80)	(-0.21)	(0.62)	(0.78)	(0.68)	(-0.91)
High ST	-0.353	-0.291	-0.289	-0.531	-0.889	-0.536***
	(-3.41)	(-2.97)	(-2.47)	(-4.29)	(-5.50)	(-3.07)
High - Low	-0.552***	-0.665***	-0.675***	-0.935***	-1.591***	-1.039***
	(-4.69)	(-5.23)	(-4.47)	(-5.28)	(-8.64)	(-6.07)

Table A2. Double sorts on salience and beta by deleting bottom 20% small stocks

This table reports the monthly average percentage alphas with respect to Fama and French (2015) 5-factor (FF-5) for portfolios sorted on stock beta (BETA) and salience theory measure (ST). The residual BETA is computed as the residual of cross-sectional regression of stock beta on idiosyncratic skewness (ISKEW). The stocks are first sorted into quintiles based on RES_BETA and then into quintiles based on ST. The portfolio is held for 1 month and equal-weight. All variables are defined in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. Bottom 20% stocks are deleted according to their market capitalization. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	Low BETA	2	3	4	High BETA	High - Low
Low ST	0.280	0.294	0.324	0.231	0.476	0.196
	(3.75)	(3.79)	(3.45)	(1.82)	(2.30)	(1.26)
2	0.160	0.130	0.106	0.230	0.185	0.025
	(2.50)	(2.02)	(1.58)	(2.82)	(1.57)	(0.18)
3	0.016	0.035	-0.011	0.140	0.070	0.054
	(0.25)	(0.56)	(-0.18)	(1.53)	(0.70)	(0.42)
4	-0.049	-0.078	-0.151	-0.081	-0.274	-0.225
	(-0.77)	(-1.28)	(-2.33)	(-1.12)	(-2.54)	(-1.60)
High ST	-0.339	-0.339	-0.426	-0.428	-0.952	-0.614***
	(-3.83)	(-4.39)	(-5.54)	(-4.31)	(-6.22)	(-3.14)
High - Low	-0.619***	-0.633***	-0.750***	-0.659***	-1.428***	-0.809***
	(-7.13)	(-6.86)	(-6.97)	(-4.35)	(-6.76)	(-3.53)

Table A3. Double sorts on salience and alternative beta estimators

This table reports the monthly average percentage FF-5 alphas for portfolios sorted on alternative stock beta and salience theory measure (ST). Stocks are first sorted into quintiles based on one of alternative betas and then into quintiles based on ST. The portfolio is held for 1 month and equal-weight. Panels A, B and C report double-sorts results using shrinkage adjustment BETA, non-synchronous adjustment BETA and Frazzini and Pedersen's (2014) BETA, respectively. Detailed variable constructions are shown in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	Low BETA	2	3	4	High BETA	High - Low
Panel A. Shrinkage adjustment BETA						
Low ST	0.364	0.406	0.456	0.388	0.817	0.452**
	(4.55)	(5.04)	(4.42)	(2.89)	(4.64)	(2.57)
2	0.135	0.156	0.147	0.277	0.235	0.100
	(2.24)	(2.25)	(2.08)	(3.00)	(1.92)	(0.71)
3	0.043	0.071	0.059	0.154	0.230	0.188
	(0.65)	(1.10)	(0.91)	(1.82)	(2.12)	(1.37)
4	0.039	0.034	-0.054	0.001	-0.163	-0.202
	(0.62)	(0.54)	(-0.78)	(0.01)	(-1.44)	(-1.40)
High ST	-0.213	-0.277	-0.376	-0.495	-1.005	-0.792***
	(-2.16)	(-3.42)	(-4.20)	(-4.36)	(-5.89)	(-4.09)
High - Low	-0.577***	-0.682***	-0.832***	-0.883***	-1.822***	-1.245***
	(-6.12)	(-6.88)	(-6.89)	(-5.78)	(-8.55)	(-5.88)
Panel B. Non-synchronous adjustment BETA						
Low ST	0.307	0.344	0.421	0.620	0.714	0.407**
	(3.94)	(4.69)	(4.39)	(4.52)	(3.99)	(2.19)
2	0.156	0.100	0.217	0.244	0.406	0.250*
	(2.56)	(1.71)	(2.95)	(2.41)	(3.05)	(1.65)
3	0.017	0.043	0.060	0.136	0.210	0.193
	(0.27)	(0.72)	(0.90)	(1.69)	(1.79)	(1.40)
4	-0.003	0.016	-0.075	-0.003	-0.097	-0.093
	(-0.05)	(0.27)	(-1.20)	(-0.04)	(-0.81)	(-0.63)
High ST	-0.207	-0.264	-0.493	-0.505	-0.933	-0.726***
	(-2.34)	(-3.47)	(-5.58)	(-4.85)	(-5.17)	(-3.66)
High - low	-0.514***	-0.608***	-0.914***	-1.125***	-1.646***	-1.133***
	(-5.54)	(-7.01)	(-8.19)	(-7.55)	(-7.48)	(-5.19)
Panel C. FP (2014) BETA						
Low ST	0.478	0.548	0.573	0.425	0.278	-0.201
	(4.54)	(5.75)	(5.46)	(3.33)	(1.60)	(-1.08)
2	0.254	0.146	0.123	0.113	0.039	-0.215
	(2.91)	(2.19)	(1.71)	(1.36)	(0.29)	(-1.30)
3	0.154	0.198	0.026	-0.004	-0.039	-0.193
	(1.94)	(3.33)	(0.42)	(-0.06)	(-0.37)	(-1.32)
4	0.301	0.101	-0.010	-0.086	-0.210	-0.511***
	(3.49)	(1.48)	(-0.16)	(-1.15)	(-1.68)	(-3.07)
High ST	0.079	-0.105	-0.383	-0.439	-0.918	-0.997***
	(0.64)	(-0.93)	(-3.81)	(-3.58)	(-5.85)	(-5.53)
High - Low	-0.399***	-0.653***	-0.956***	-0.864***	-1.196***	-0.797***
	(-3.11)	(-5.10)	(-6.70)	(-4.88)	(-5.98)	(-4.17)

Table A4. Double sorts on salience and residual beta

This table reports the monthly average percentage alphas with respect to Fama and French (2015) 5-factor (FF-5) for portfolios sorted on stock residual beta (RES_BETA) and salience theory measure (ST). The residual BETA is computed as the residual of cross-sectional regression of stock beta on idiosyncratic skewness (ISKEW). The stocks are first sorted into quintiles based on RES_BETA and then into quintiles based on ST. The portfolio is held for 1 month and equal-weight. All variables are defined in the appendix. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	Low BETA	2	3	4	High BETA	High - Low
Low RES_ST	0.284	0.368	0.461	0.411	0.751	0.467***
	(3.54)	(4.64)	(4.54)	(3.07)	(4.17)	(2.59)
2	0.129	0.107	0.099	0.225	0.257	0.127
	(2.08)	(1.59)	(1.36)	(2.36)	(2.06)	(0.88)
3	0.025	0.080	0.045	0.164	0.187	0.162
	(0.39)	(1.30)	(0.66)	(1.92)	(1.74)	(1.21)
4	0.057	0.057	-0.045	-0.097	-0.161	-0.218
	(0.93)	(0.90)	(-0.67)	(-1.24)	(-1.41)	(-1.53)
High RES_ST	-0.211	-0.300	-0.270	-0.454	-1.018	-0.807***
	(-2.08)	(-3.60)	(-2.76)	(-3.76)	(-6.06)	(-4.27)
High - Low	-0.496***	-0.669***	-0.731***	-0.866***	-1.769***	-1.273***
	(-4.84)	(-6.67)	(-6.25)	(-5.06)	(-8.15)	(-6.07)

Table A5. Single sort on alternative risk measures

This table reports the monthly average percentage alphas with respect to Fama and French (2015) 5-factor (FF-5) for portfolios sorted on one of risk proxies. TVOL is the standard deviation of returns over the month. CVOL is the cash flow volatility computed as the standard deviation of cash flow from operations over the past 5 years. COV is the analyst coverage as the number of analysts following the firm in the last year. ESKEW is expected skewness. The variables are defined in the appendix. Risk quintiles are formed based on one of the risk proxies in month *t*. The portfolio is held for 1 month in month *t*+1 and equal-weight. High risk firms are those with high TVOL, and high CVOL, low COV and high ESKEW. High - Low refers to the high minus low risk portfolios. The sample includes NYSE/AMEX/NASDAQ common stocks with a price of at least $1 per share and excludes financial firms and firms with negative book equity. The sample period is from July 1963 to December 2018. All *t*-statistics (in parentheses) are calculated using Newey-West (1987) standard errors and asterisks refer to the level of significance: *** (1%), ** (5%), * (10%). For brevity, we only report asterisks for the hedged portfolios.

	TVOL	CVOL	COV	ESKEW
Low risk	0.024	0.065	-0.018	0.024
	(0.44)	(1.37)	(-0.20)	(0.38)
2	0.150	0.082	-0.062	0.037
	(2.84)	(1.56)	(-0.74)	(0.67)
3	0.235	0.088	-0.133	0.008
	(4.03)	(1.59)	(-1.77)	(0.12)
4	0.139	0.099	-0.087	0.100
	(1.82)	(1.49)	(-1.02)	(0.84)
High risk	-0.381	0.047	0.006	0.075
	(-2.97)	(0.48)	(0.06)	(0.35)
High - Low	-0.405***	-0.018	0.024	0.050
	(-2.87)	(-0.17)	(0.21)	(0.23)